

KMT-2026-BLG-0521 / OGLE-2026-BLG-0058

The restricted-set reductions (v4 and v5) — standalone report

Prepared 2026-09-08. Solutions: `~/KMTdata/Results/v4b/IFfinal_<field>.mat` and `~/KMTdata/Results/v5/IFfinal_<field>.mat` (variable IFsys in each). This document describes **only** these two runs, which share one source set and differ in exactly one respect: v4 pins the frame to Gaia during the fit, v5 fits everything freely and ties to Gaia afterwards. The decade reduction that remains the project's reference solution is described in `KMT260058_astrometry_report.pdf`.

1. What this run is

A reduction built to a different specification from the reference one, to test two ideas: restrict the analysis to a clean calibration set, and tie the frame to Gaia from the start rather than afterwards.

1. **Analysis set.** $14 < I_{\text{OGLE}} < 17$, no companion brighter than $I = 18$ within 2.0 arcsec, and a Gaia counterpart with **RUWE** < 1.4 . The target is added explicitly — at $I = 18.13$ it falls outside the magnitude window. Every other source is **dropped at the input**, so all measurements, detrending and quality cuts act on this set alone. 2. **The frame is pinned to Gaia.** Each calibration star's position *and* proper motion are held at Gaia's values, converted into cut-out pixels. The per-epoch affine is therefore fitted against Gaia positions propagated to that epoch by Gaia's own motions. The target is left free; pinning it would erase the signal. The six-parameter gauge is gone by construction and no post-hoc tie is needed. 3. **Passengers.** Three stars at the target's own magnitude ($I \sim 18$), outside the calibration set, are carried along unpinned so that something comparable to the target can be plotted in the same frame.

	BLG41	BLG01
raw sources in the MSc file	1177	1187
surviving the standard quality cuts — the reference reduction's set	594	621
analysis set of these runs	130	100
of which pinned to Gaia (v4 only)	126	96
free: target + passengers	4	4
epochs	19 133	19 981

The row that matters for comparison is the second: the reference reduction fits **594 and 621** sources, and these runs fit 130 and 100 of them. The raw file holds about 1180 entries, but half of those are marginal detections that the standard conversion discards before any reduction sees them — they are found in a median of **10%** of epochs, against 80% for the sources that survive. Every source detected in more than half the epochs is kept, without exception. So the restriction imposed here is 130 out of 594, not 130 out of 1177.

2. The calibration set

Selection, all conditions required:

- $14 < I_{\text{OGLE}} < 17$

- **no companion brighter than $I = 18$ within 2.0 arcsec** (5 KMT pixels at 0.4"/pix), the companion list being the OGLE catalogue, which resolves about six times more sources than KMT detects
- **a Gaia DR3 counterpart with RUWE < 1.4**

	BLG41	BLG01
in the magnitude window, isolated	144	127
with a Gaia counterpart	144	116
after RUWE < 1.4	126	96

The RUWE cut removes 12% and 17% — consistent with the 87% pass rate for $G < 17$ across the whole cone, so crowding is not biasing it heavily.

3. Gaia

RUWE is not in the local catsHTM DR3 table. It carries `astrometric_chi2_al` and `astrometric_n_good_obs_al` but neither RUWE nor the `u0(G, BP-RP)` table needed to normalise UWE into it. The true values were queried from the Gaia archive instead — 9464 sources in a 3 arcmin cone, cached in `~/KMTdata/GaiaRef/`. This mattered: at $G < 17$, **87%** pass RUWE < 1.4, where the unnormalised UWE proxy predicted 74%, so the proxy would have discarded about 13% of good stars for no reason.

Gaia's proper-motion errors for these stars are **0.070 / 0.044 mas/yr**, some fifty times better than our internal frame. That is what makes pinning worthwhile in principle.

The Gaia-to-pixel map is fitted directly from the matched pairs, not carried through OGLE. The OGLE bridge is needed only to identify which star is which; routing coordinates through it inherits OGLE's ~110 mas astrometric error.

	through OGLE	fitted directly
BLG41, median offset / scatter	252 / 159 mas	23 / 16 mas
BLG01	351 / 148 mas	21 / 20 mas

The fitted scale is 2.519 (BLG41) and 2.529 (BLG01) pix/arcsec against the 2.500 expected at 0.4"/pix — an independent check that the map is right. The remaining ~20 mas is static per star, so it displaces each star's residuals by a constant and does not affect proper motion or any time-dependent signal.

4. The chain

With proper motions supplied by Gaia, the joint proper-motion fit and the per-season fits of the reference reduction are unnecessary. Two steps remain:

1. **Full-decade SysRem correction**, frame pinned, 2 components on the astrometric residuals. 2. **Final solution** reusing that correction, with the pixel-phase term on, 15 weighted iterations before SysRem and 6 after.

Convergence over the frame stars, last three iterations: **0.0009 mas (BLG41)** and **0.0026 mas (BLG01)**.

5. Results

	BLG41	BLG01
frame-star residual RMS, $\Delta X / \Delta Y$	6.13 / 5.20 mas	5.30 / 5.42 mas
target residual RMS, $\Delta X / \Delta Y$	29.62 / 22.63 mas	28.73 / 23.25 mas

For comparison, the reference reduction gives bright-star 6.41 / 6.88 and 6.48 / 7.01, and target **24.62 / 21.00** and **25.48 / 21.78**.

The calibration stars are measured better and the target worse. The stars are bright and isolated, which is what they were selected for; the target is **14 to 20% worse** than in the reference reduction. The cause is the frame: it is now built from 126 stars rather than ~600 weighted sources, which roughly doubles its ideal noise and raises its leverage by a factor of twenty to fifty. Restricting the analysis set is what costs. The Gaia tie is not at fault.

Superseded — absolute proper motions. Every Gaia tie made before the v6 reduction removed only the **constant** part of the gauge. Relative astrometry leaves the proper motions free up to a constant *plus a linear gradient across the field*, and the residual gradient is a shear of about -0.036 mas/yr per pixel, found at nearly the same value in both fields. Removing the full six-parameter gauge cuts the scatter of our motions about Gaia from 4.34 / 3.57 to 0.98 / 0.66 mas/yr (BLG41) and 3.48 / 3.20 to 1.00 / 0.70 (BLG01), and moves the target's motion by a few tenths of a mas/yr. **All absolute proper motions in this document are therefore superseded by KMT260058_v6_report.pdf**, which gives $-2.017 \pm 0.335 / -7.163 \pm 0.360$ (BLG41) and $-1.781 \pm 0.256 / -7.136 \pm 0.377$ (BLG01) mas/yr. Relative motions, residual scatters and everything else here are unaffected.

6. Proper motions, and whether the two fields agree

Converting the target's motion out of pixel axes with each field's own map:

	BLG41	BLG01	difference
v4, Gaia-pinned	-1.799 / -6.780	-1.158 / -7.147	0.640 / 0.367
reference, post-hoc tie	-2.92 / -7.25	-2.22 / -7.24	0.702 / 0.004

mas/yr, as $(\mu_\alpha \cos \delta, \mu_\delta)$. Both agree at about the 1 mas/yr level, which is the target's own uncertainty, but **pinning the frame did not tighten the agreement**. The post-hoc tie was already doing this job.

The passengers agree better than the target does. Three stars at the target's magnitude, freely fitted, carried through both fields:

star	BLG41	BLG01	difference
$l = 18.14$, 4.5 pix from the target	-0.63 / -5.47	-1.08 / -5.47	0.45 / 0.00
$l = 17.91$, 21 pix	-6.81 / -9.99	-6.94 / -10.06	0.13 / 0.07
$l = 18.05$, 48 pix	+3.06 / -6.34	+2.80 / -6.48	0.26 / 0.14
the target , $l = 18.13$	-1.80 / -6.78	-1.16 / -7.15	0.64 / 0.37

The two clean passengers reproduce two to five times better than the target. The 4.5-pix star's apparent agreement is not a good sign — see section 7.

Sensitivity to the set. Adding only those three passengers to a set of 127 moved the fitted target motion by **0.42 / 0.63**

mas/yr on BLG41 and 0.09 / 0.41 on BLG01. A result that moves by half a mas per year when three faint stars join the set is not a precise result, and this is the most direct measurement here of how uncertain the target's motion actually is.

7. A blended neighbour tracks the event

The passenger 4.5 pixels from the target behaves unlike anything else in the run. Comparing the 2026 season against the 2016–2022 baseline:

star	distance	ΔX (BLG01 / BLG41)	ΔY (BLG01 / BLG41)	season scatter
target	0	+2.7 / +2.6	+7.1 / +6.4	3.6–5.3
passenger 1	4.5 pix	–44.1 / –57.8	–27.9 / –32.2	16–19
passenger 2	21 pix	+5.6 / +3.5	+2.9 / +3.3	2.3–3.7
passenger 3	48 pix	–3.4 / –5.5	–2.5 / –2.1	2.1–5.3

mas. The excursion is **40 to 58 mas at 4.5 pixels and gone by 21 pixels**, reproduced independently in both fields, and that star's season-to-season scatter is four times the others' in every season, not only in 2026. With a seeing FWHM near 7 pixels, a star 4.5 pixels away shares most of its light with the target: as the target brightens through the event, the blend's centroid moves. The neighbour moves *away* from the target, consistently in both fields, which is what PSF fitting a blend tends to do as flux is reattributed to the brightening component.

Coordinates. Pixel positions are per field, since the two cut-outs are offset by about 1.1 pixels; sky positions are derived by inverting each field's own Gaia-to-pixel map, and the two fields agree on them to **2 to 9 mas**, which is an independent check on both maps.

star	I_OGLE	dist [pix]	BLG41 pixel	BLG01 pixel	RA (J2000)	Dec (J2000)
target	18.13	0.0	150.4, 150.4	151.6, 150.2	17:52:38.085	–31:47:36.21
cal_d24	16.68	24.5	161.3, 172.2	162.4, 172.1	17:52:37.746	–31:47:27.53
cal_d30	16.63	30.5	120.4, 154.6	121.4, 154.5	17:52:39.023	–31:47:34.60
cal_d43	16.94	42.9	110.9, 166.6	111.9, 166.5	17:52:39.318	–31:47:29.89
out_d04	18.14	4.5	146.4, 148.3	147.6, 148.0	17:52:38.210	–31:47:37.06
out_d21	17.91	21.3	155.3, 171.1	156.4, 170.9	17:52:37.935	–31:47:28.01
out_d48	18.05	48.5	198.6, 145.4	199.8, 145.0	17:52:36.585	–31:47:38.10

The target's recovered position, 17:52:38.085 –31:47:36.21, sits 0.11 arcsec from the OGLE EWS coordinates the pipeline was pointed at, which is the accuracy of the centroid plus the map, not a disagreement.

Three consequences:

- **That star is unusable as a comparison.** It was chosen as the closest at the target's magnitude, and being closest is exactly what disqualifies it. Passengers 2 and 3 are the meaningful ones.
- **Its good inter-field agreement in section 6 is misleading.** Both fields see the same blending artefact from the same event, so agreeing is not evidence of a good measurement.
- **It bears directly on the target.** The blend is mutual. If the neighbour's centroid moves 40 to 58 mas when the target brightens, the target's centroid is being pulled by the same shared flux. The target's own +2.7 / +7.1 mas

in 2026 therefore cannot be read cleanly as astrometric microlensing; some or all of it may be blending. That is a systematic at the tens-of-mas level against a predicted signal of a few mas.

8. Overlap of the two fields

The two cut-outs cover the same sky, so in principle the same stars should appear in both. Counting how many actually do is complicated by blending: a close pair may be split into two detections in one field and merged into one in the other, and matching the two source lists directly then counts the same star twice. Stars are therefore identified **through the OGLE catalogue**, which is at 0.26 arcsec/pix and resolves about six times more sources than KMT detects. Each KMT source is matched to an OGLE entry within 1.5 pix, and two sources sharing an entry are the same star.

	BLG41	BLG01
sources	594	621
matched to an OGLE entry	551 (93%)	497 (80%)
distinct OGLE stars behind them	454	394

The gap between the last two rows is the blending, measured directly: 551 detections correspond to only 454 stars on BLG41, so about **97 detections are duplicates** — one star split into two — and 103 on BLG01.

OGLE I	BLG41	BLG01	in both	distinct	found in both
< 15	28	22	22	28	78.6%
15 – 16	54	42	42	54	77.8%
16 – 17	172	149	146	175	83.4%
17 – 18	135	119	117	137	85.4%
18 – 19	55	44	41	58	70.7%
all brighter than 19	444	376	368	452	81.4%

So **about four stars in five are detected in both fields**, and the shortfall is not a faint-end effect: the fraction is flat at 78 to 85% from the brightest stars down to I = 18, falling only in the last magnitude. What limits it is that BLG01 simply detects fewer stars — 376 against 444 brighter than 19 — which traces back to its poorer OGLE match rate, 80% against 93%.

There is nothing fainter than I = 19 to compare. Our matched source lists span 12.87 to 18.98 in both fields, while the OGLE catalogue itself reaches 21.40 and holds 2660 entries fainter than 19. That is the KMT detection limit in this crowding, not a catalogue cut, so no faint population exists to match.

An earlier version of this section matched the two source lists directly and reported 92 and 93% agreement. That was **one-way nearest-neighbour matching**, which lets two sources in one field both claim the same source in the other and counts both as matched — precisely the blend-splitting case. Requiring the match to be mutual collapses 546 one-way matches to 453 real pairs, and saturates there at any radius beyond 1.5 pix. The OGLE-identified figure above is the one to use.

9. v5 — the same set, freely fitted

v4 changes two things at once against the reference reduction: it shrinks the source set, and it pins the frame. v5 separates them. It uses **exactly the same source list** as v4b — read from the same file, not re-selected — the same two steps, the same iteration counts and the same SysRem and pixel-phase settings. The only difference is that nothing is pinned: every calibration star is freely fitted and takes part in the frame determination at every step, and Gaia enters only afterwards, as a post-hoc tie.

So **v3 to v5 measures what restricting the set costs**, and **v5 to v4b measures what pinning does**.

Target residual RMS

		BLG41 $\Delta X / \Delta Y$	BLG01 $\Delta X / \Delta Y$
v3 — ~600 sources, free, post-hoc tie		24.62 / 21.00	25.48 / 21.78
v5 — 130/100 sources, free, post-hoc tie		25.61 / 22.27	28.45 / 23.15
v4b — the same sources, pinned to Gaia		29.62 / 22.63	28.73 / 23.25

step	isolates	BLG41 X / Y	BLG01 X / Y
v3 → v5	restricting the set	+4.0% / +6.0%	+11.7% / +6.3%
v5 → v4b	pinning the frame	+15.7% / +1.6%	+1.0% / +0.4%

Both hurt and neither dominates consistently. Restricting the set costs 4 to 12% everywhere; pinning costs almost nothing except in BLG41's X, where it costs 16%. The reference reduction remains the best for the target on every axis in both fields.

Frame stars. Over exactly the same 126 / 96 stars:

	BLG41	BLG01
v5, freely fitted	4.87 / 4.81	4.77 / 4.77
v4b, pinned to Gaia	6.13 / 5.20	5.30 / 5.42

4.8 mas is the best frame-star astrometry this project has produced. The comparison is not entirely fair — the free fit adjusts four parameters per star and so absorbs part of each star's own systematics, which the pinned version cannot — but the ~1.3 mas gap is the price of holding stars at catalogue values.

Convergence over the frame stars: 0.0041 mas (BLG41) and 0.0002 (BLG01).

Inter-field agreement of the target's absolute motion.

	difference between the fields
v3	0.702 / 0.004
v5, <code>gaiaTie</code>	0.072 / 0.123
v5, positional-map tie	0.363 / 0.294
v4b	0.640 / 0.367

mas/yr. v5 agrees best, but **the headline 0.072 should not be quoted alone**. `gaiaTie` fits a 2 by 2 matrix from a proper-motion regression, and on these small sets it returned rotations of **146.34 deg for BLG41 and 108.65 for BLG01** — 38 deg apart, where the direct positional map puts the two cut-outs 0.126 deg apart. That matrix is fitting

noise. Re-tying independently, with the linear part taken from the positional map and only the gauge constant from the proper-motion comparison, gives 0.363 / 0.294 instead. The honest statement is that v5's fields agree to **0.07 to 0.36 in RA and 0.12 to 0.29 in Dec**, and that the choice of tie method is itself a systematic of 0.2 to 0.6 mas/yr.

The gauge removed from v5 is $-0.219 / +4.758$ mas/yr (BLG41, 130 calibrators) and $+2.834 / +3.983$ (BLG01, 100). With it removed, the seven plotted stars agree between the fields to 0.25 to 0.78 mas/yr in RA and 0.02 to 0.87 in Dec.

10. Verdict

Neither v4 nor v5 is adopted as the primary solution. Restricting the analysis to the calibration set costs 4 to 12% on the target, and pinning the frame to Gaia costs a further 0 to 16%; the reference reduction is better on every axis in both fields. What the pair does establish, by differencing, is that **both choices cost something and the set restriction is the more consistent penalty** — pinning is nearly free except on BLG41's X axis.

v5 nevertheless produces the best frame stars (4.8 mas) and the best inter-field agreement of the target's absolute motion of any reduction here, and it does so while leaving every calibration star free to take part in the frame. If a restricted-set reduction is wanted, v5 rather than v4 is the one to use.

Its most valuable output is not the astrometry but section 7: the direct measurement of how far a blended neighbour is dragged by the event, which applies to the target in every reduction we have made.

11. Figures and files

file	content
report_v4_RMS.png	residual RMS against OGLE I, per axis and field, for this run
report_v4_motion_<field>_target.png	the target, sidereal-month bins, 2x2
report_v4_motion_<field>_cal_d{24,30,43}.png	three calibration stars, proper motion pinned to Gaia
report_v4_motion_<field>_out_d{04,21,48}.png	the three I ~ 18 passengers, freely fitted
report_v5_motion_<field>_*.png	the same seven stars in v5 , where nothing is pinned; the quoted sky motion is absolute

The `_d NN` suffix is the star's distance from the target in pixels, and names the **same physical star in both fields** — the figures are directly comparable across fields. (An earlier version ranked the comparison stars per field, so `cal1` denoted different objects in BLG41 and BLG01.)

Axis convention. The Gaia-to-pixel map has a negative determinant: it contains a parity flip, and the pixel X axis runs *opposite* to increasing RA.

	rotation, flip removed	scale
BLG41	-0.317 deg	2.5194 pix/arcsec
BLG01	-0.444 deg	2.5290 pix/arcsec

So the cut-outs are essentially aligned with RA and Dec apart from that X flip, with a third of a degree of residual rotation, and the two fields differ in orientation by only 0.126 deg. In each figure the **legend** quotes the motion along the

pixel axis and the **panel heading** quotes it on the sky, so the two carry opposite signs in X by construction; both are labelled accordingly.

In every motion figure the columns are the two axes, X left and Y right; the top row keeps the proper motion and the bottom row removes it. **A pinned star's red line is Gaia's proper motion, not a fitted one**, so its lower panels test the frame, whereas the target's and the passengers' test the fit.

Figures

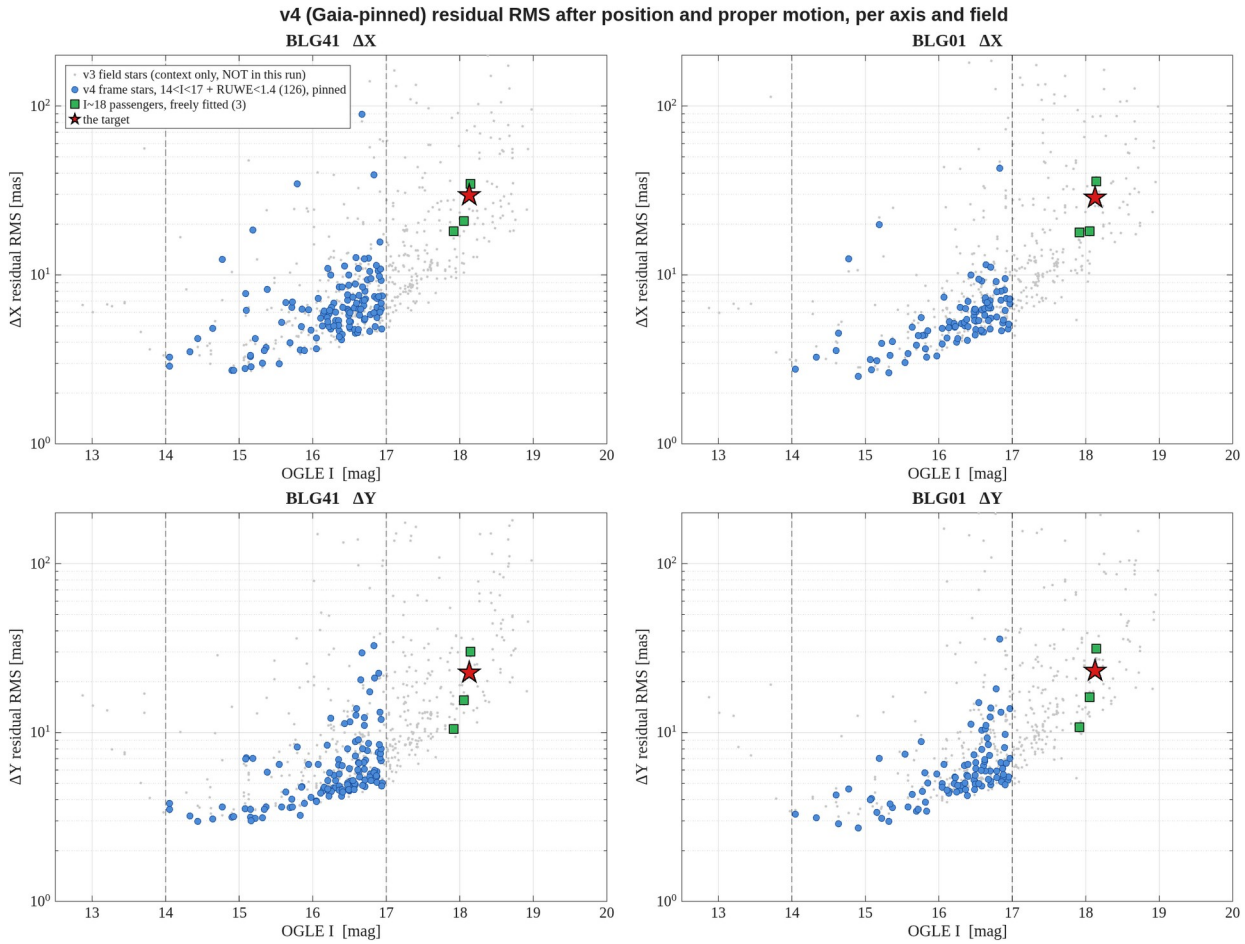


Figure 1. Residual RMS against the OGLE catalogue magnitude, per axis and per field, **for this run**, after position and proper motion have been removed. Blue: the 126 / 96 frame stars, $14 < I < 17$ with $\text{RUWE} < 1.4$, pinned to Gaia. Green: the three 1-18 passengers, freely fitted. Red: the target. Grey: field stars from the reference reduction, shown for context only — **they are not part of this run**. Dashed lines mark the 14 and 17 mag limits.

report_v4_RMS.png

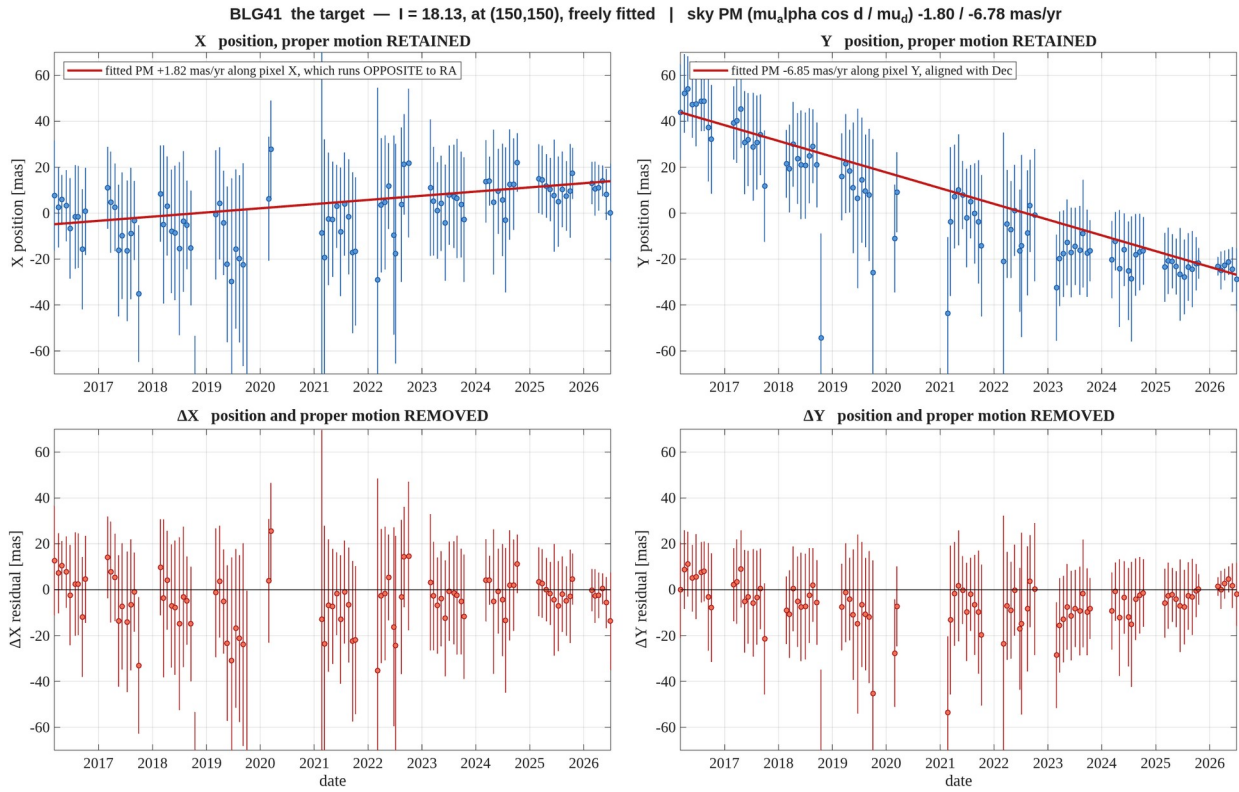


Figure 2. **BLG41** — the target, freely fitted. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.** The legend quotes the motion along the pixel axis and the heading quotes it on the sky; **pixel X runs opposite to RA**, so those two carry opposite signs in X.

report_v4_motion_BLG41_target.png

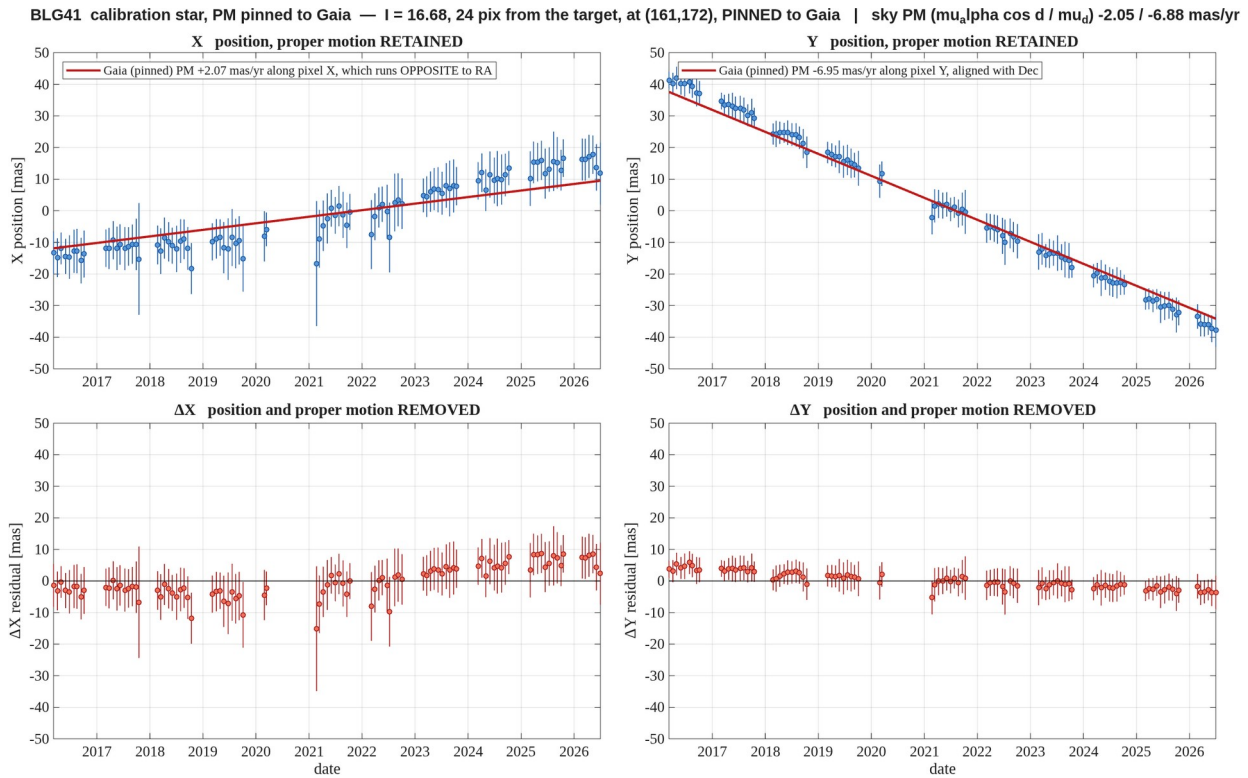


Figure 3. **BLG41** — calibration star 24 pix from the target, proper motion pinned to Gaia. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.** The legend quotes the motion along the pixel axis and the heading quotes it on the sky; **pixel X runs opposite to RA**, so those two carry opposite signs in X.
report_v4_motion_BLG41_cal_d24.png

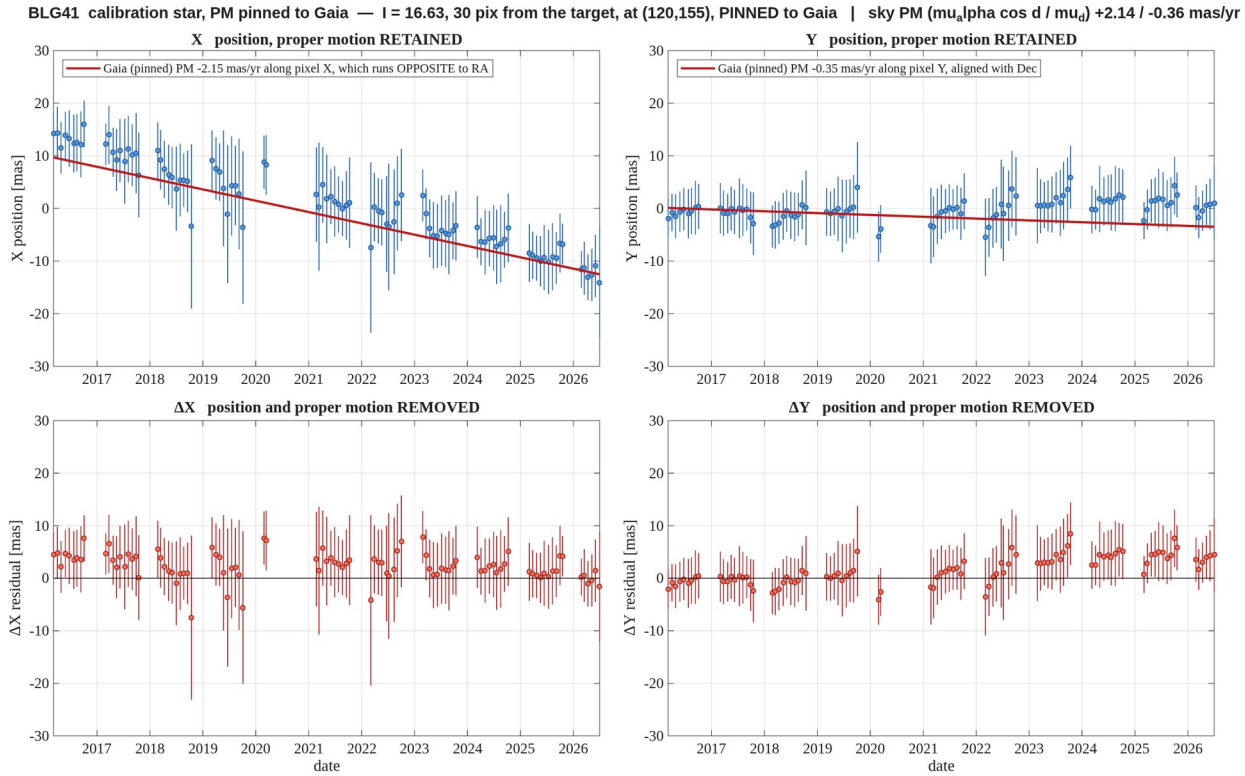


Figure 4. **BLG41** — calibration star 30 pix from the target, proper motion pinned to Gaia. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.** The legend quotes the motion along the pixel axis and the heading quotes it on the sky; **pixel X runs opposite to RA**, so those two carry opposite signs in X.
report_v4_motion_BLG41_cal_d30.png

BLG41 calibration star, PM pinned to Gaia — $I = 16.94$, 43 pix from the target, at (111,167), PINNED to Gaia | sky PM ($\mu_\alpha \cos d / \mu_\delta$) $+0.23 / -7.85$ mas/yr

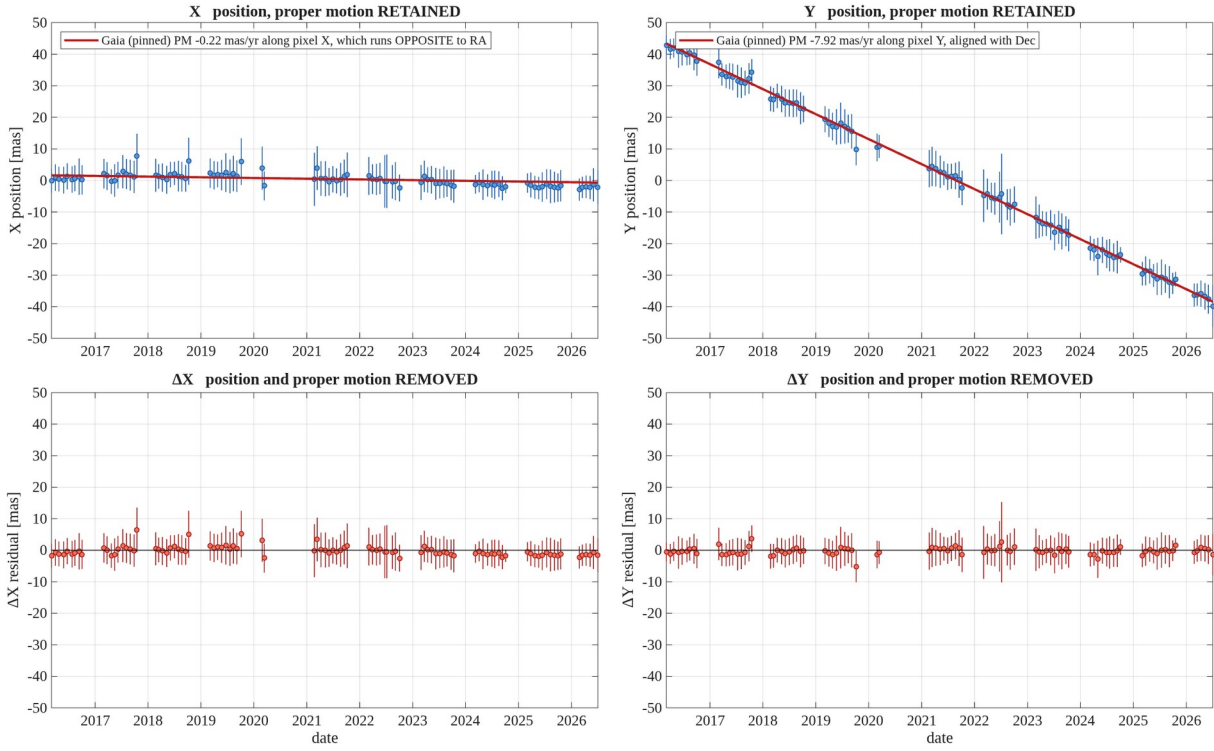


Figure 5. **BLG41** — calibration star 43 pix from the target, proper motion pinned to Gaia. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.** The legend quotes the motion along the pixel axis and the heading quotes it on the sky; **pixel X runs opposite to RA**, so those two carry opposite signs in X.

report_v4_motion_BLG41_cal_d43.png

BLG41 comparison star outside the set, PM fitted — $I = 18.14$, 5 pix from the target, at (146,148), freely fitted | sky PM ($\mu_\alpha \cos d / \mu_\delta$) $-0.63 / -5.47$ mas/yr

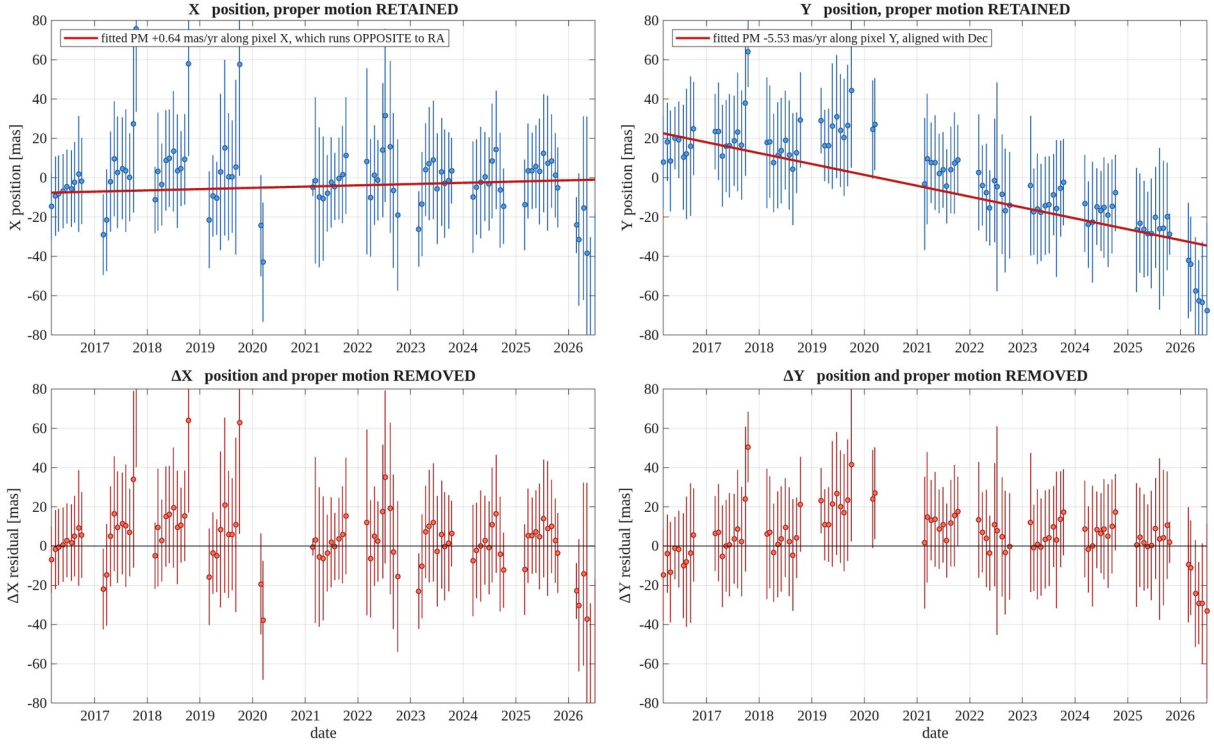


Figure 6. **BLG41** — passenger at the target magnitude, 4.5 pix away — **blended with the target**, see section 7. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.** The legend quotes the motion along the pixel axis and the heading quotes it on the sky; **pixel X runs opposite to RA**, so those two carry opposite signs in X.
report_v4_motion_BLG41_out_d04.png

BLG41 comparison star outside the set, PM fitted — $I = 17.91$, 21 pix from the target, at (155,171), freely fitted | sky PM ($\mu_\alpha \lvert \text{pha} \cos d / \mu_\delta$) $-6.81 / -9.99$ mas/yr

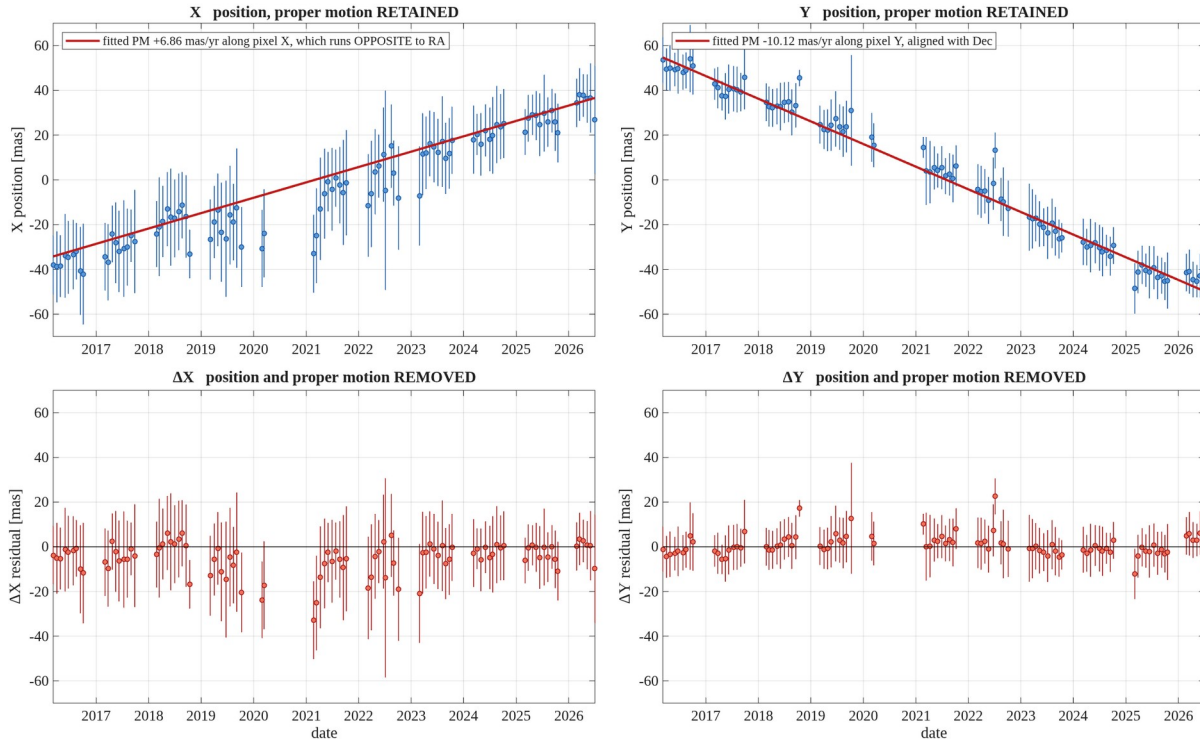


Figure 7. **BLG41** — passenger at the target magnitude, 21 pix away, freely fitted. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.** The legend quotes the motion along the pixel axis and the heading quotes it on the sky; **pixel X runs opposite to RA**, so those two carry opposite signs in X.
report_v4_motion_BLG41_out_d21.png

BLG41 comparison star outside the set, PM fitted — $I = 18.05$, 48 pix from the target, at (199,145), freely fitted | sky PM ($\mu_\alpha \cos d / \mu_\delta$) +3.06 / -6.34 mas/yr

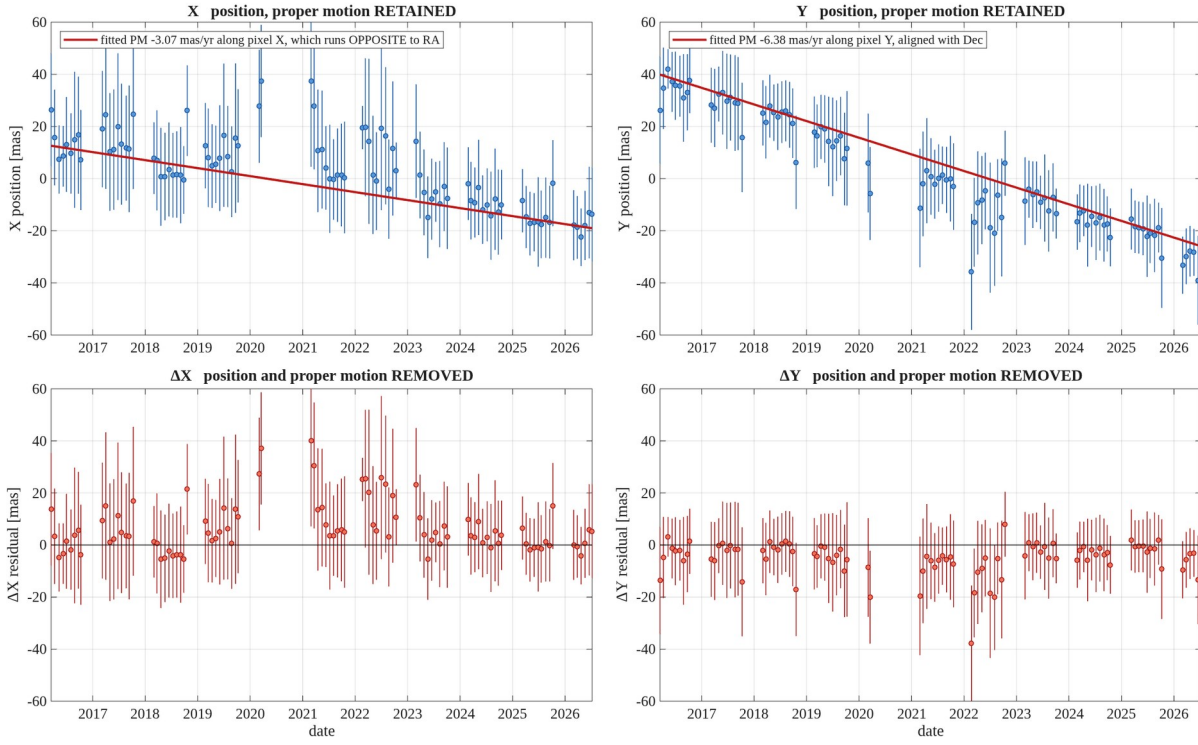


Figure 8. **BLG41** — passenger at the target magnitude, 48 pix away, freely fitted. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.** The legend quotes the motion along the pixel axis and the heading quotes it on the sky; **pixel X runs opposite to RA**, so those two carry opposite signs in X.

report_v4_motion_BLG41_out_d48.png

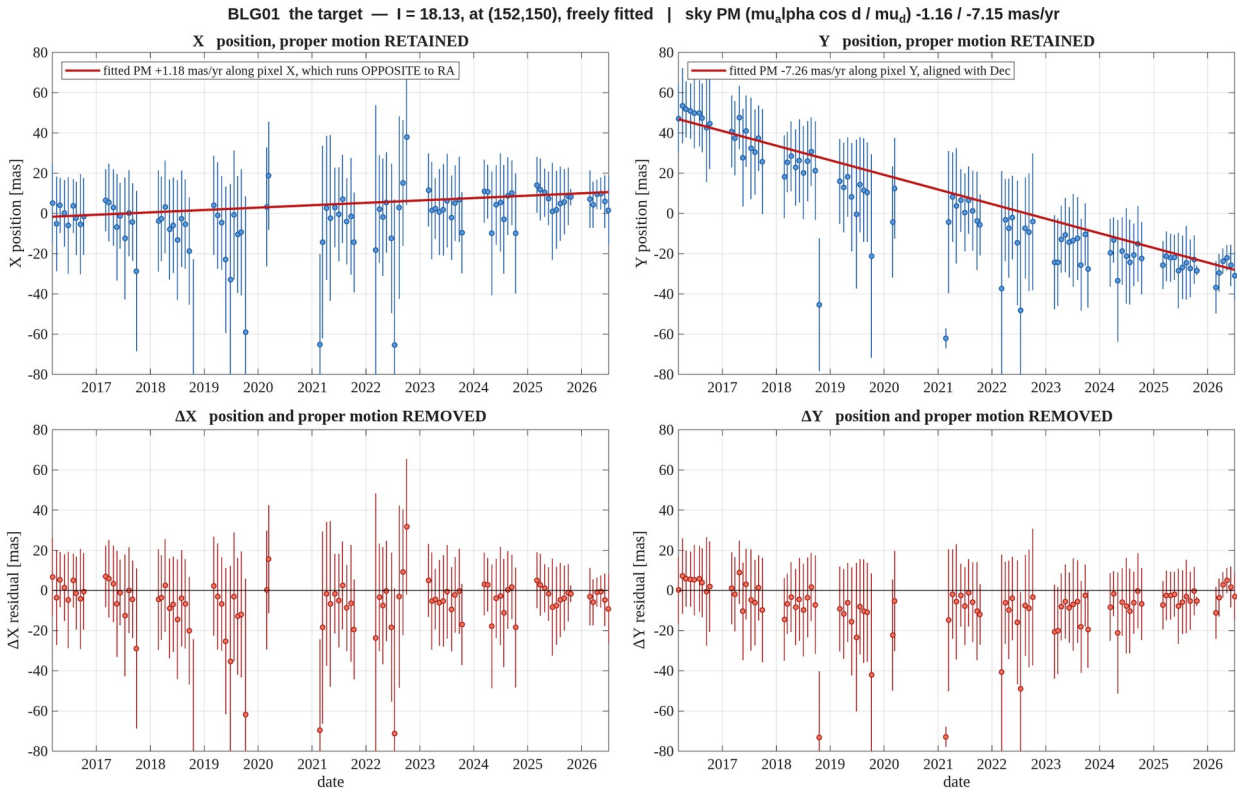


Figure 9. **BLG01** — the target, freely fitted. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.** The legend quotes the motion along the pixel axis and the heading quotes it on the sky; **pixel X runs opposite to RA**, so those two carry opposite signs in X.

report_v4_motion_BLG01_target.png

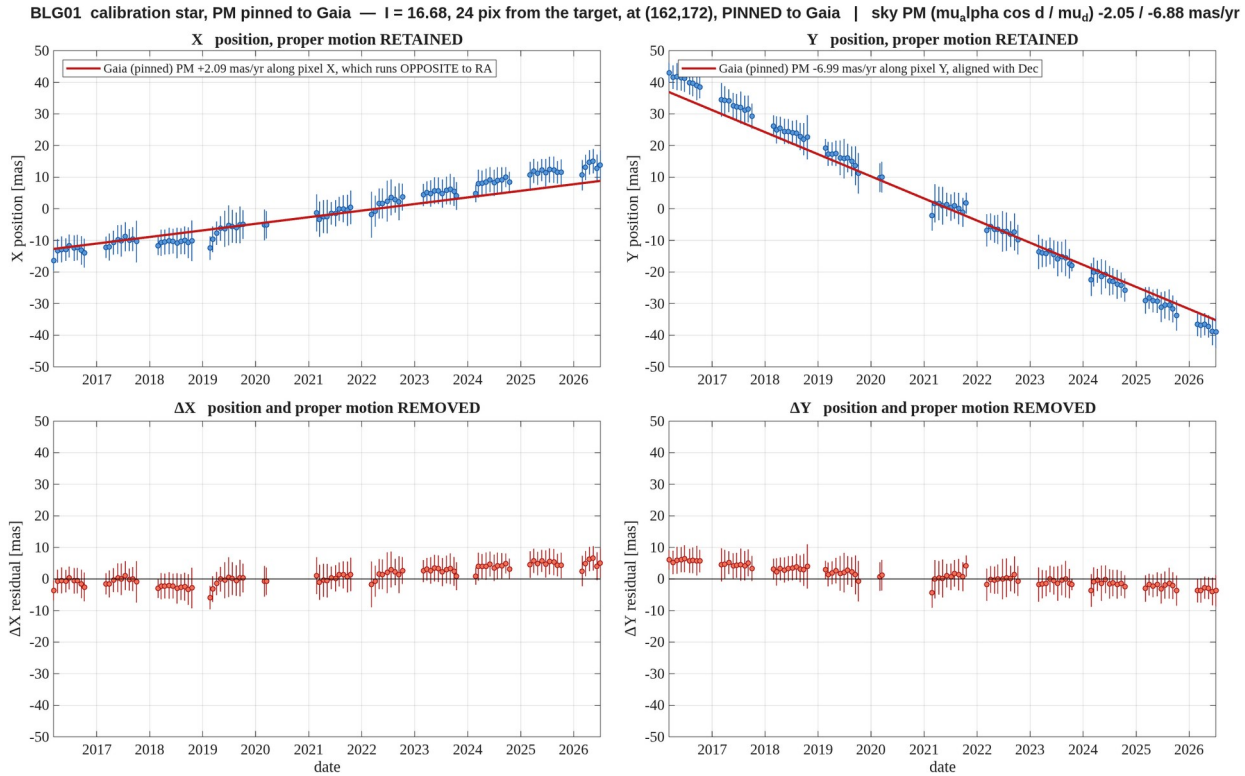


Figure 10. **BLG01** — calibration star 24 pix from the target, proper motion pinned to Gaia. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.** The legend quotes the motion along the pixel axis and the heading quotes it on the sky; **pixel X runs opposite to RA**, so those two carry opposite signs in X.

report_v4_motion_BLG01_cal_d24.png

BLG01 calibration star, PM pinned to Gaia — $I = 16.63$, 30 pix from the target, at (121,154), PINNED to Gaia | sky PM ($\mu_\alpha \cos d / \mu_d$) $+2.14 / -0.36$ mas/yr

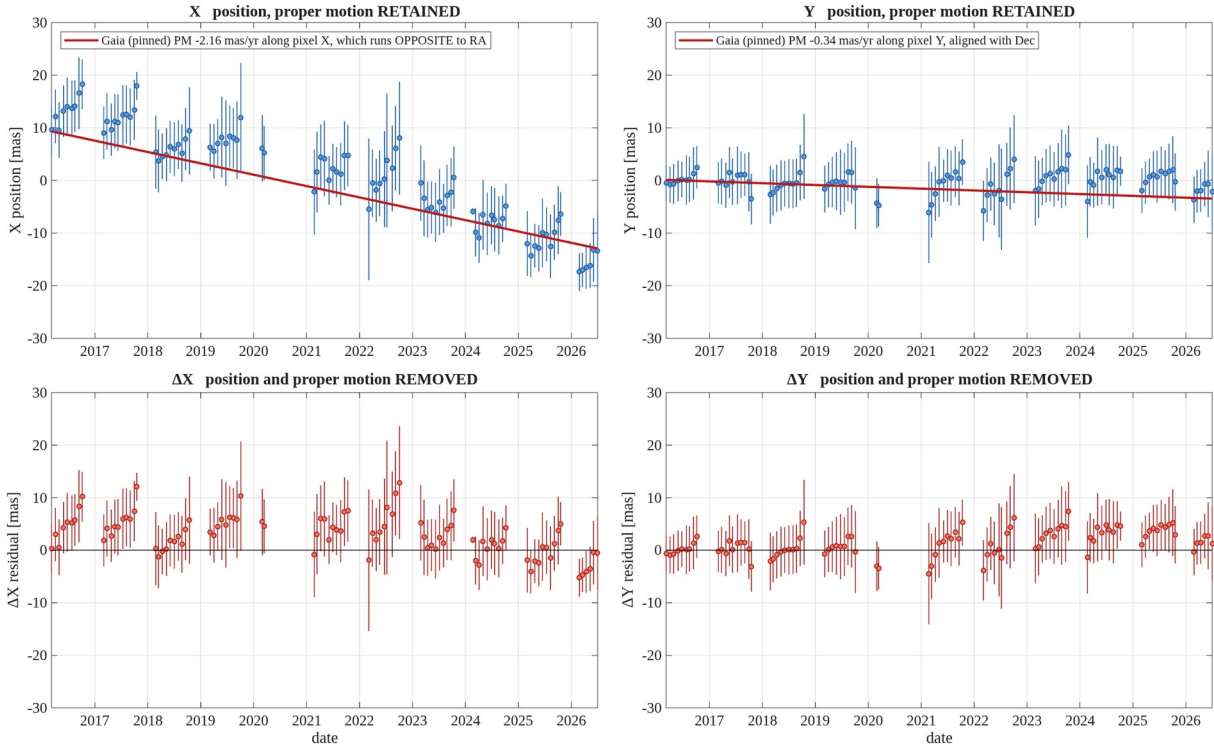


Figure 11. **BLG01** — calibration star 30 pix from the target, proper motion pinned to Gaia. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.** The legend quotes the motion along the pixel axis and the heading quotes it on the sky; **pixel X runs opposite to RA**, so those two carry opposite signs in X.

report_v4_motion_BLG01_cal_d30.png

BLG01 calibration star, PM pinned to Gaia — $I = 16.94$, 43 pix from the target, at (112,166), PINNED to Gaia | sky PM ($\mu_\alpha \cos d / \mu_d$) $+0.23 / -7.85$ mas/yr

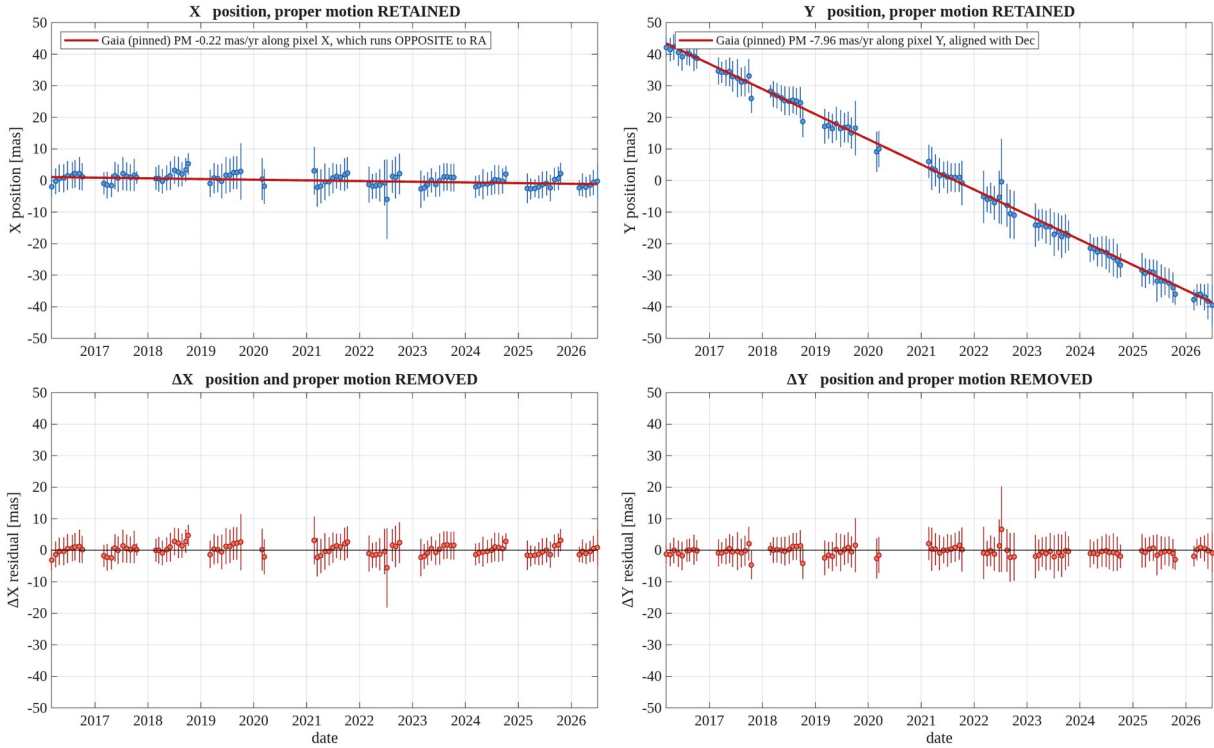


Figure 12. **BLG01** — calibration star 43 pix from the target, proper motion pinned to Gaia. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.** The legend quotes the motion along the pixel axis and the heading quotes it on the sky; **pixel X runs opposite to RA**, so those two carry opposite signs in X.
report_v4_motion_BLG01_cal_d43.png

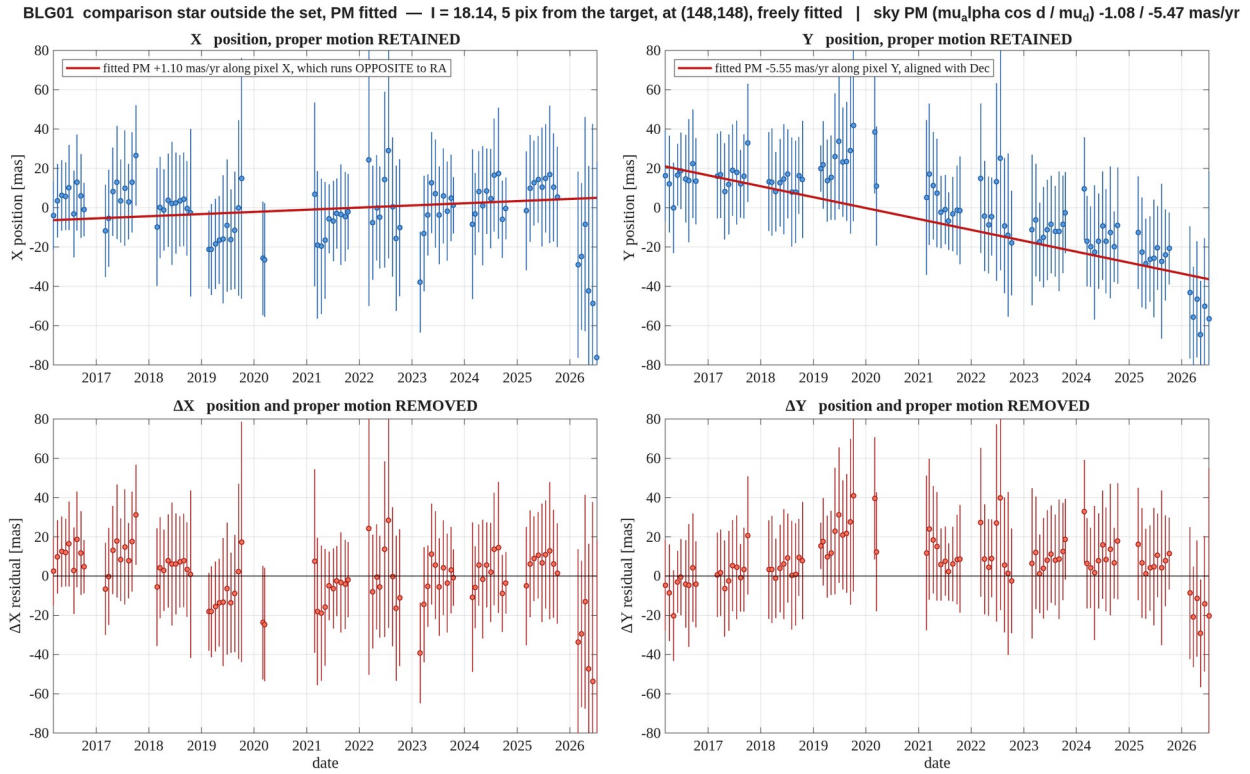


Figure 13. **BLG01** — passenger at the target magnitude, 4.5 pix away — **blended with the target**, see section 7. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.** The legend quotes the motion along the pixel axis and the heading quotes it on the sky; **pixel X runs opposite to RA**, so those two carry opposite signs in X.
report_v4_motion_BLG01_out_d04.png

BLG01 comparison star outside the set, PM fitted — $l = 17.91$, 21 pix from the target, at (156,171), freely fitted | sky PM ($\mu_\alpha \cos \delta / \mu_\delta$) -6.94 / -10.06 mas/yr

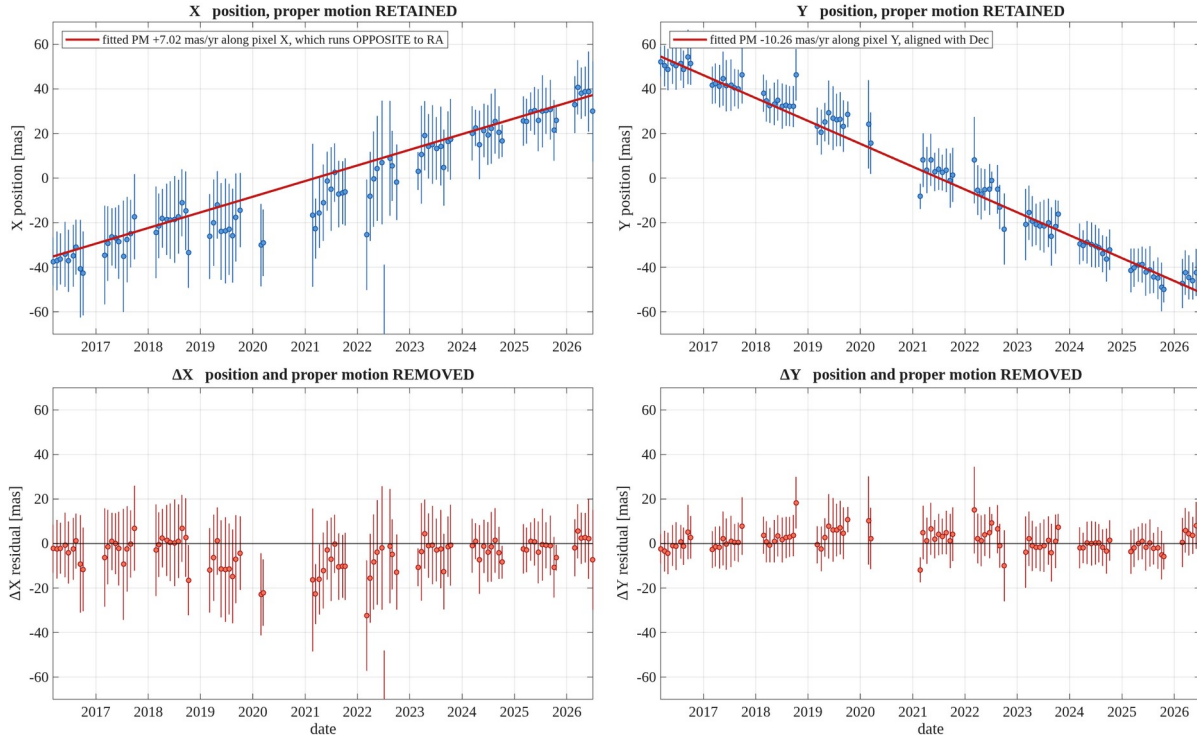


Figure 14. **BLG01** — passenger at the target magnitude, 21 pix away, freely fitted. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.** The legend quotes the motion along the pixel axis and the heading quotes it on the sky; **pixel X runs opposite to RA**, so those two carry opposite signs in X.
report_v4_motion_BLG01_out_d21.png

BLG01 comparison star outside the set, PM fitted — $l = 18.05$, 49 pix from the target, at (200,145), freely fitted | sky PM ($\mu_\alpha \cos \delta / \mu_\delta$) +2.80 / -6.48 mas/yr

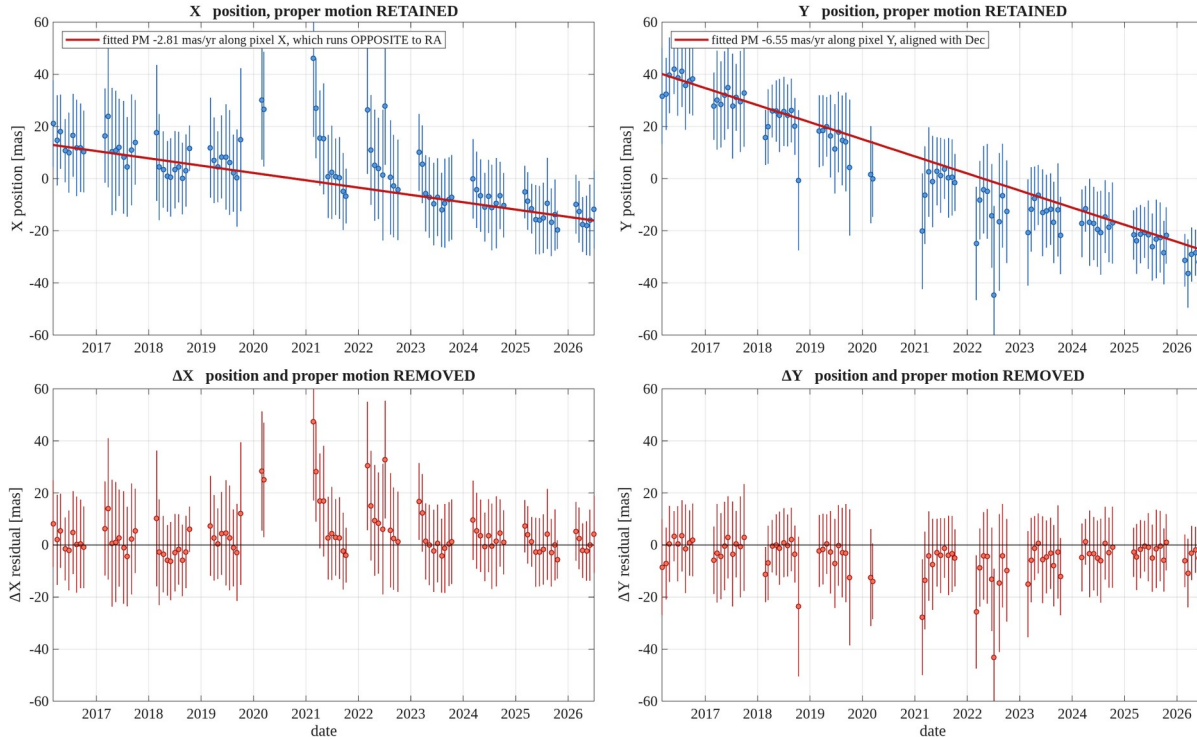


Figure 15. **BLG01** — passenger at the target magnitude, 48 pix away, freely fitted. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.** The legend quotes the motion along the pixel axis and the heading quotes it on the sky; **pixel X runs opposite to RA**, so those two carry opposite signs in X.
report_v4_motion_BLG01_out_d48.png

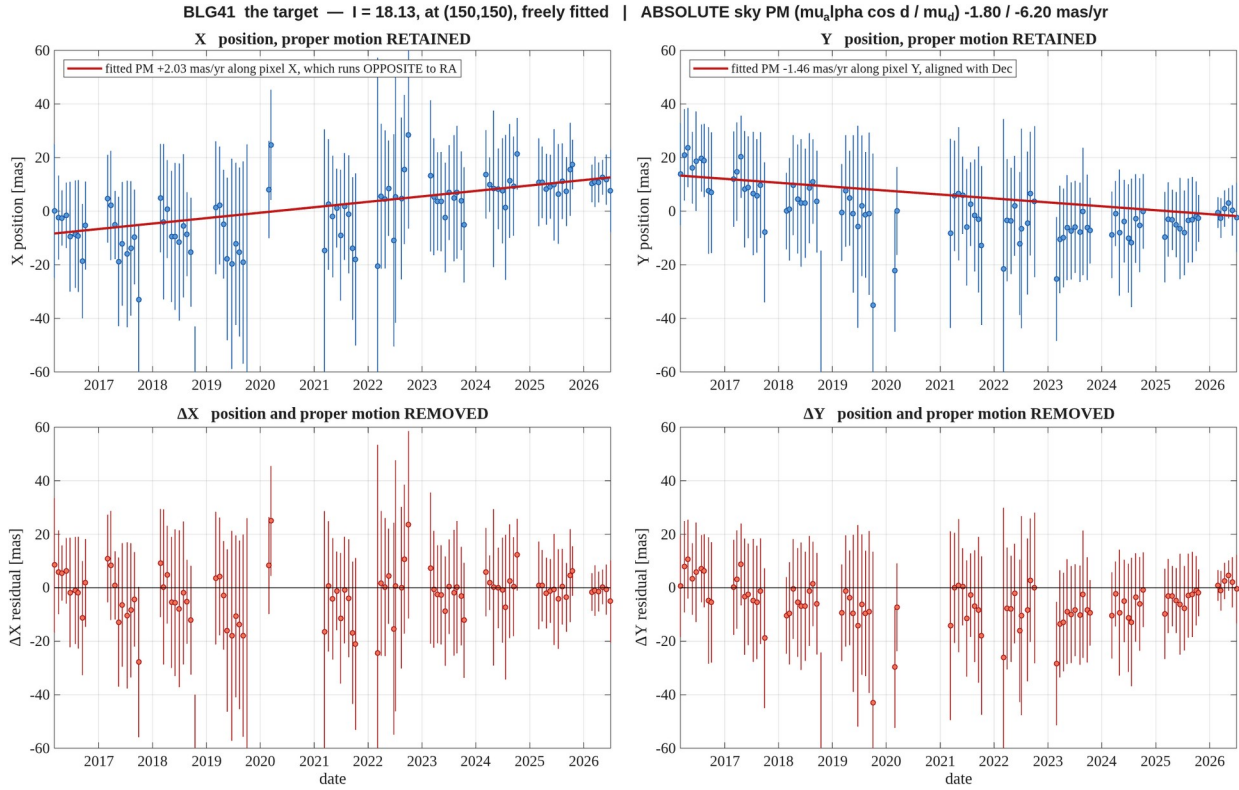


Figure 16. **v5, BLG41** — the target, freely fitted. The same star as the corresponding v4 figure, but from the run in which **nothing is pinned** and Gaia enters only as a post-hoc tie. Sky motions quoted in the heading have the gauge removed and are absolute. Binned in one sidereal month; columns are the two axes, X left and Y right; top row keeps the proper motion, bottom row removes it. **Pixel X runs opposite to RA**, so the legend and the heading carry opposite signs in X.

report_v5_motion_BLG41_target.png

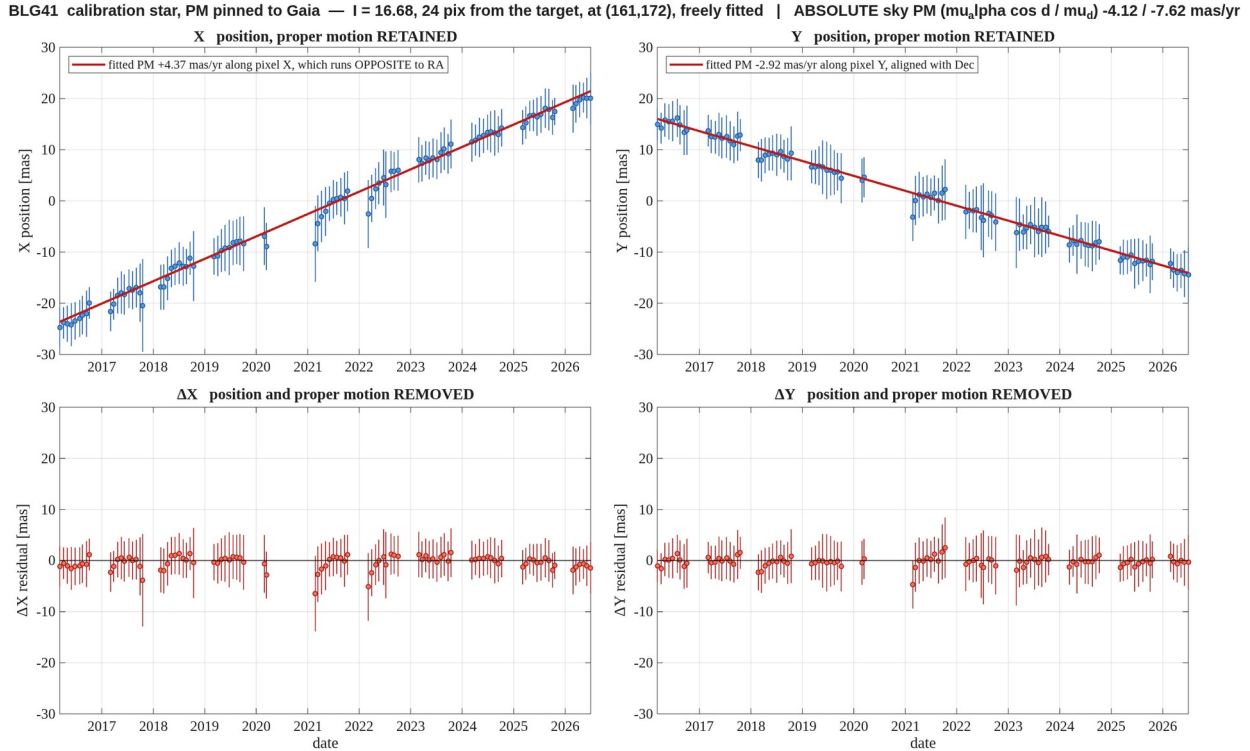


Figure 17. **v5, BLG41** — calibration star 24 pix from the target, freely fitted (v5 pins nothing). The same star as the corresponding v4 figure, but from the run in which **nothing is pinned** and Gaia enters only as a post-hoc tie. Sky motions quoted in the heading have the gauge removed and are absolute. Binned in one sidereal month; columns are the two axes, X left and Y right; top row keeps the proper motion, bottom row removes it. **Pixel X runs opposite to RA**, so the legend and the heading carry opposite signs in X.

report_v5_motion_BLG41_cal_d24.png

BLG41 calibration star, PM pinned to Gaia — $l = 16.63$, 30 pix from the target, at (120,155), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \lvert \cos d / \mu_\delta$) $+2.20 / +1.95$ mas/yr

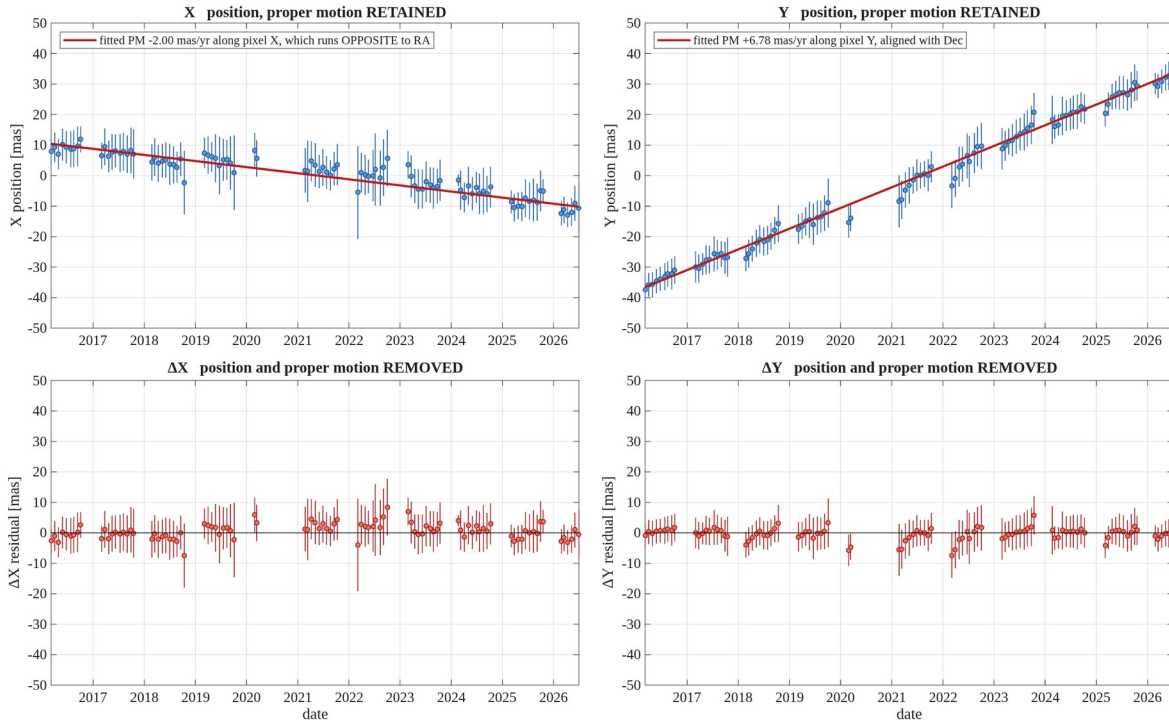


Figure 18. **v5, BLG41** — calibration star 30 pix from the target, freely fitted (v5 pins nothing). The same star as the corresponding v4 figure, but from the run in which **nothing is pinned** and Gaia enters only as a post-hoc tie. Sky motions quoted in the heading have the gauge removed and are absolute. Binned in one sidereal month; columns are the two axes, X left and Y right; top row keeps the proper motion, bottom row removes it. **Pixel X runs opposite to RA**, so the legend and the heading carry opposite signs in X. `report_v5_motion_BLG41_cal_d30.png`

BLG41 calibration star, PM pinned to Gaia — $l = 16.94$, 43 pix from the target, at (111,167), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \lvert \cos d / \mu_\delta$) $-0.28 / -5.96$ mas/yr

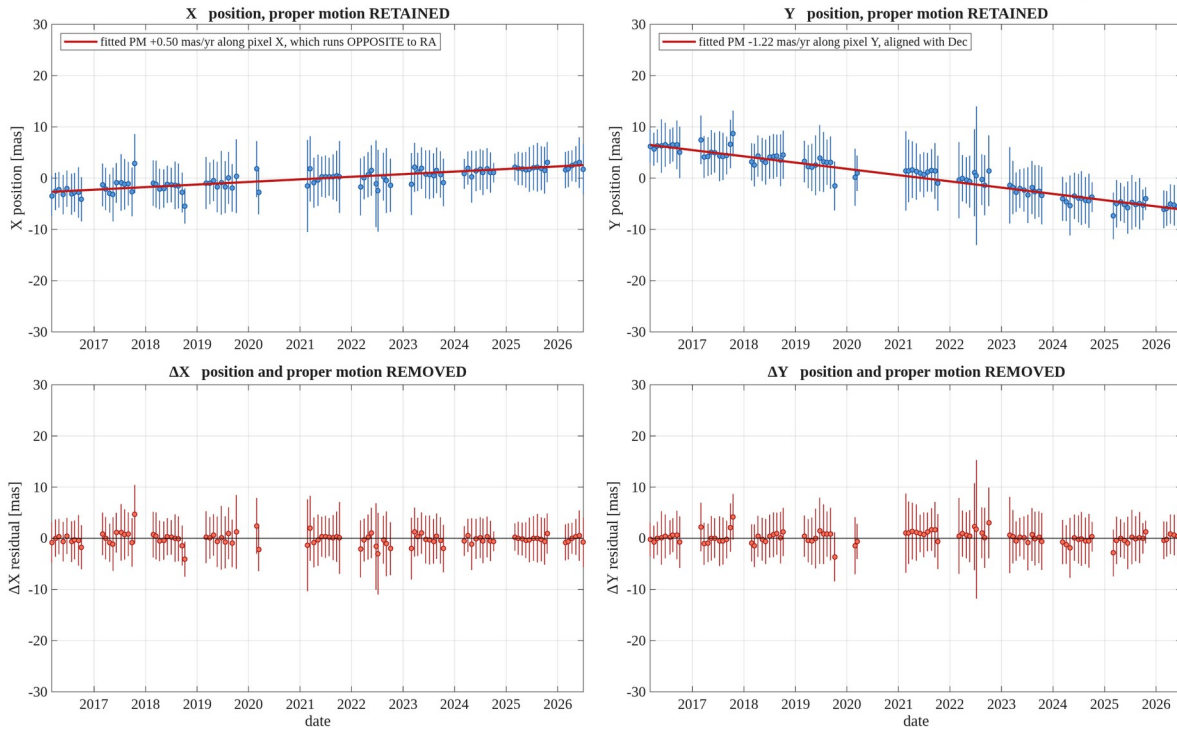


Figure 19. **v5, BLG41** — calibration star 43 pix from the target, freely fitted (v5 pins nothing). The same star as the corresponding v4 figure, but from the run in which **nothing is pinned** and Gaia enters only as a post-hoc tie. Sky motions quoted in the heading have the gauge removed and are absolute. Binned in one sidereal month; columns are the two axes, X left and Y right; top row keeps the proper motion, bottom row removes it. **Pixel X runs opposite to RA**, so the legend and the heading carry opposite signs in X. `report_v5_motion_BLG41_cal_d43.png`

BLG41 comparison star outside the set, PM fitted — $l = 18.14$, 5 pix from the target, at (146,148), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \cos \delta / \mu_\delta$) $-0.17 / -4.55$ mas/y

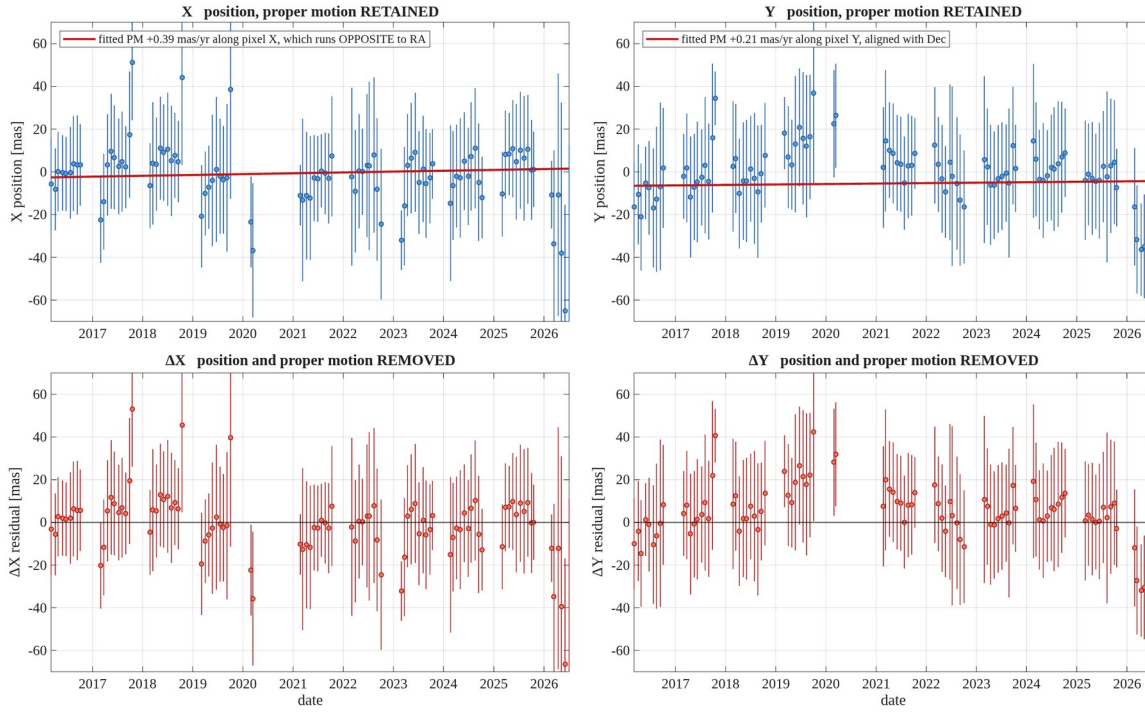


Figure 20. **v5, BLG41** — passenger at the target magnitude, 4.5 pix away — **blended with the target**, see section 7. The same star as the corresponding v4 figure, but from the run in which **nothing is pinned** and Gaia enters only as a post-hoc tie. Sky motions quoted in the heading have the gauge removed and are absolute. Binned in one sidereal month; columns are the two axes, X left and Y right; top row keeps the proper motion, bottom row removes it. **Pixel X runs opposite to RA**, so the legend and the heading carry opposite signs in X.

report_v5_motion_BLG41_out_d04.png

BLG41 comparison star outside the set, PM fitted — $l = 17.91$, 21 pix from the target, at (155,171), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \cos \delta / \mu_\delta$) $-7.71 / -9.55$ mas/y

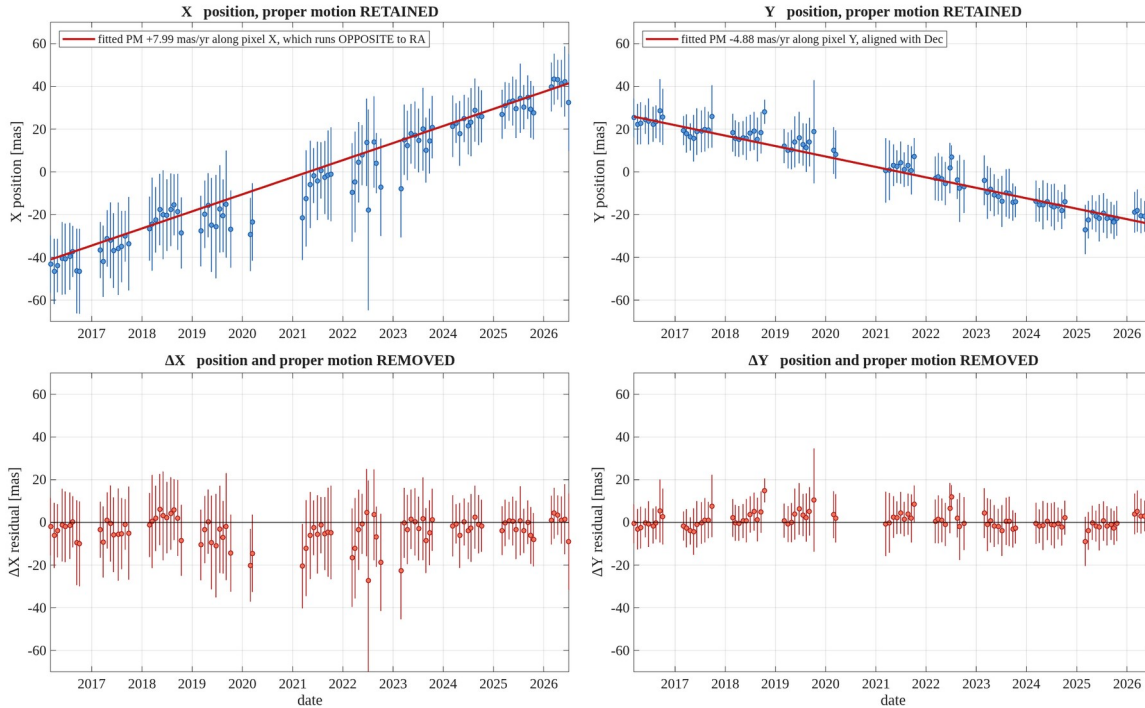


Figure 21. **v5, BLG41** — passenger at the target magnitude, 21 pix away, freely fitted. The same star as the corresponding v4 figure, but from the run in which **nothing is pinned** and Gaia enters only as a post-hoc tie. Sky motions quoted in the heading have the gauge removed and are absolute. Binned in one sidereal month; columns are the two axes, X left and Y right; top row keeps the proper motion, bottom row removes it. **Pixel X runs opposite to RA**, so the legend and the heading carry opposite signs in X.

report_v5_motion_BLG41_out_d21.png

LG41 comparison star outside the set, PM fitted — $l = 18.05$, 48 pix from the target, at (199,145), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \cos d / \mu_\delta$) +3.08 / -7.56 mas/

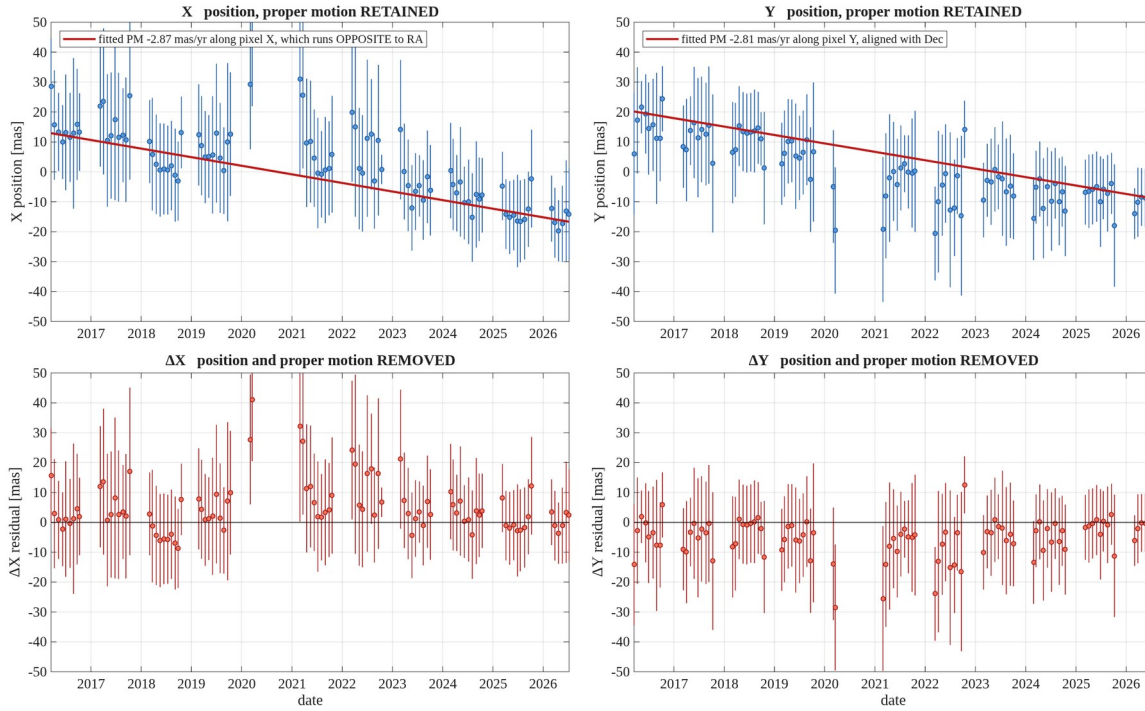


Figure 22. **v5, BLG41** — passenger at the target magnitude, 48 pix away, freely fitted. The same star as the corresponding v4 figure, but from the run in which **nothing is pinned** and Gaia enters only as a post-hoc tie. Sky motions quoted in the heading have the gauge removed and are absolute. Binned in one sidereal month; columns are the two axes, X left and Y right; top row keeps the proper motion, bottom row removes it. **Pixel X runs opposite to RA**, so the legend and the heading carry opposite signs in X. report_v5_motion_BLG41_out_d48.png

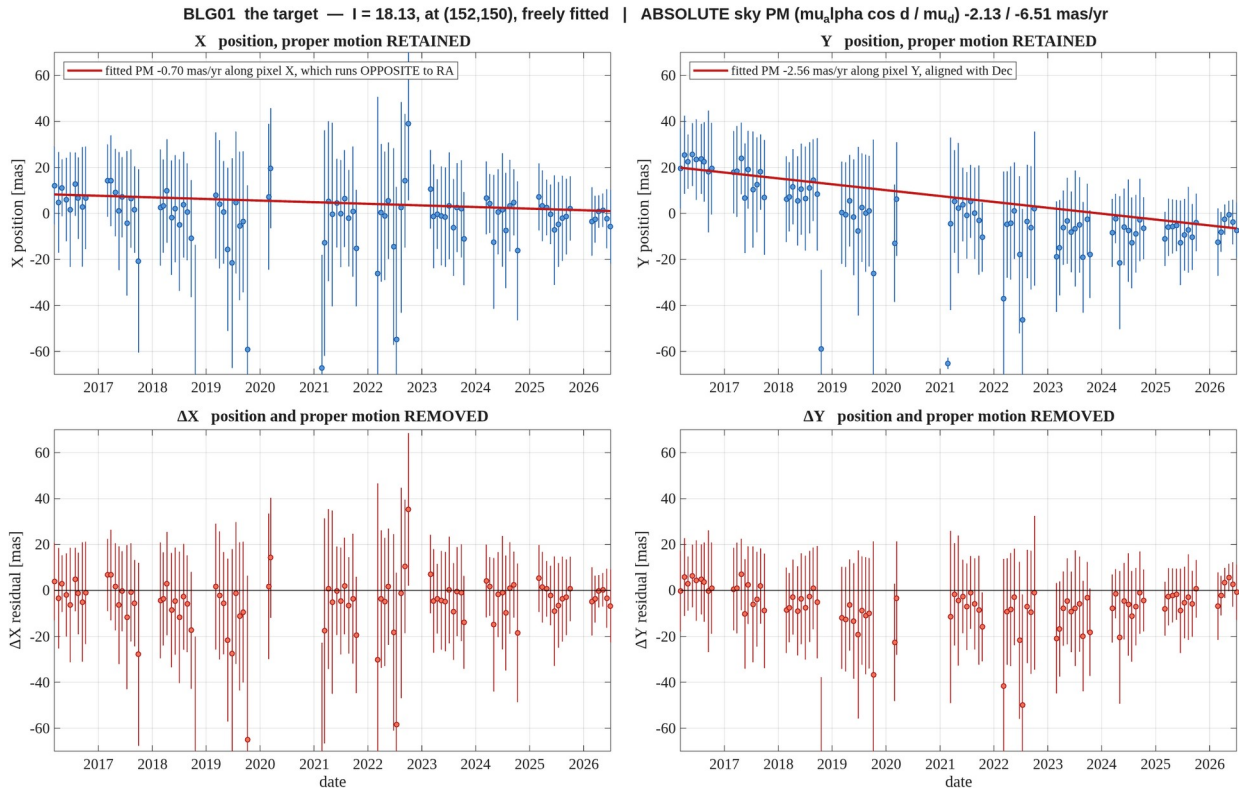


Figure 23. **v5, BLG01** — the target, freely fitted. The same star as the corresponding v4 figure, but from the run in which **nothing is pinned** and Gaia enters only as a post-hoc tie. Sky motions quoted in the heading have the gauge removed and are absolute. Binned in one sidereal month; columns are the two axes, X left and Y right; top row keeps the proper motion, bottom row removes it. **Pixel X runs opposite to RA**, so the legend and the heading carry opposite signs in X. report_v5_motion_BLG01_target.png

BLG01 calibration star, PM pinned to Gaia — $l = 16.68$, 24 pix from the target, at (162,172), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \cos d / \mu_\delta$) -4.59 / -7.73 mas/yr

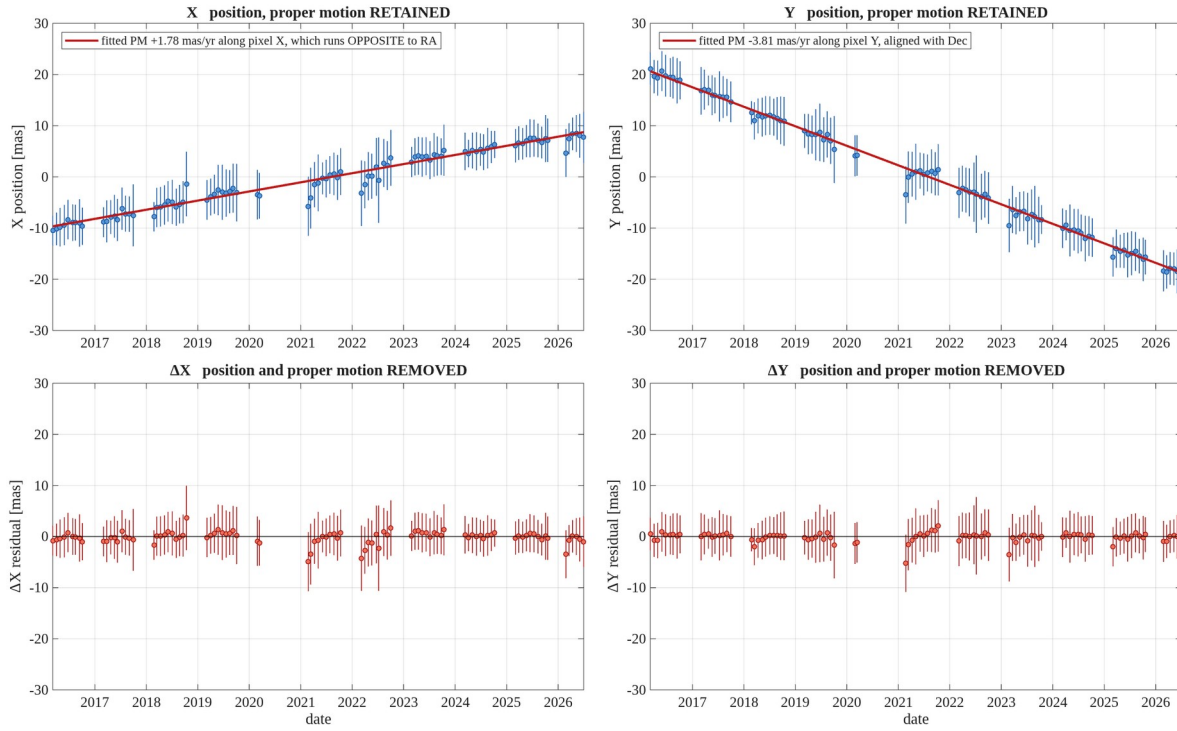


Figure 24. **v5, BLG01** — calibration star 24 pix from the target, freely fitted (v5 pins nothing). The same star as the corresponding v4 figure, but from the run in which **nothing is pinned** and Gaia enters only as a post-hoc tie. Sky motions quoted in the heading have the gauge removed and are absolute. Binned in one sidereal month; columns are the two axes, X left and Y right; top row keeps the proper motion, bottom row removes it. **Pixel X runs opposite to RA**, so the legend and the heading carry opposite signs in X.
report_v5_motion_BLG01_cal_d24.png

BLG01 calibration star, PM pinned to Gaia — $l = 16.63$, 30 pix from the target, at (121,154), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \cos d / \mu_\delta$) +1.58 / +1.97 mas/yr

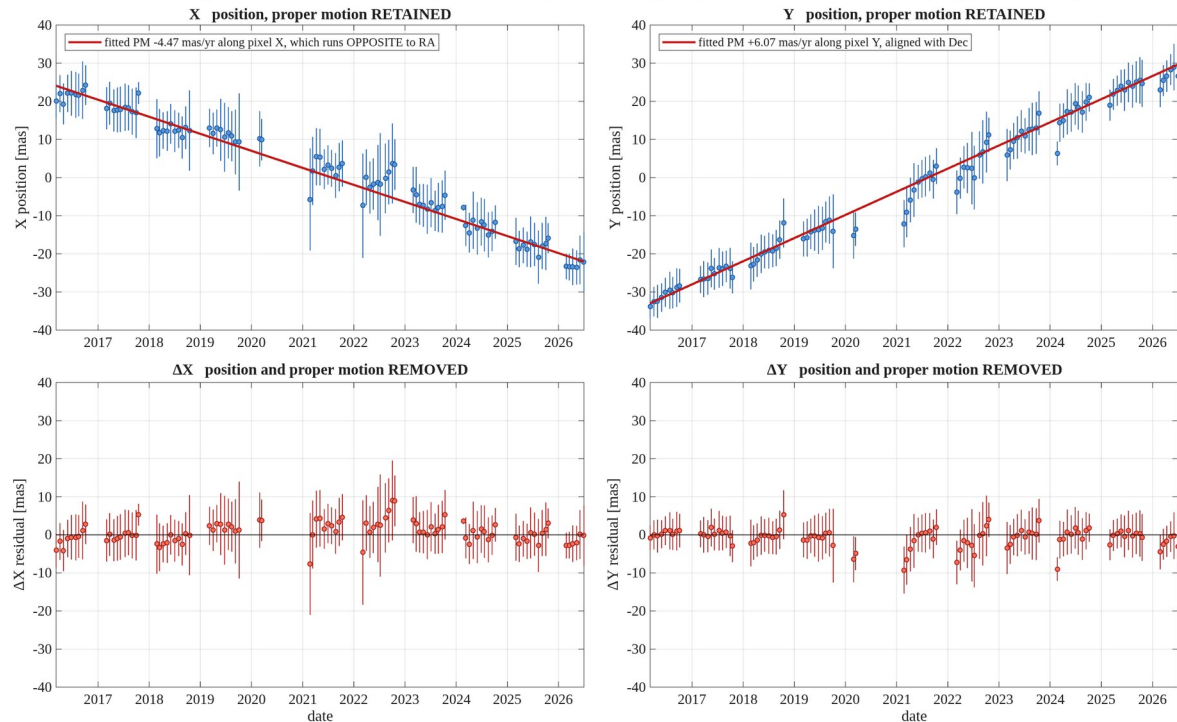


Figure 25. **v5, BLG01** — calibration star 30 pix from the target, freely fitted (v5 pins nothing). The same star as the corresponding v4 figure, but from the run in which **nothing is pinned** and Gaia enters only as a post-hoc tie. Sky motions quoted in the heading have the gauge removed and are absolute. Binned in one sidereal month; columns are the two axes, X left and Y right; top row keeps the proper motion, bottom row removes it. **Pixel X runs opposite to RA**, so the legend and the heading carry opposite signs in X.
report_v5_motion_BLG01_cal_d30.png

BLG01 calibration star, PM pinned to Gaia — $l = 16.94$, 43 pix from the target, at (112,166), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \cos d / \mu_\delta$) $-1.00 / -5.99$ mas/yr

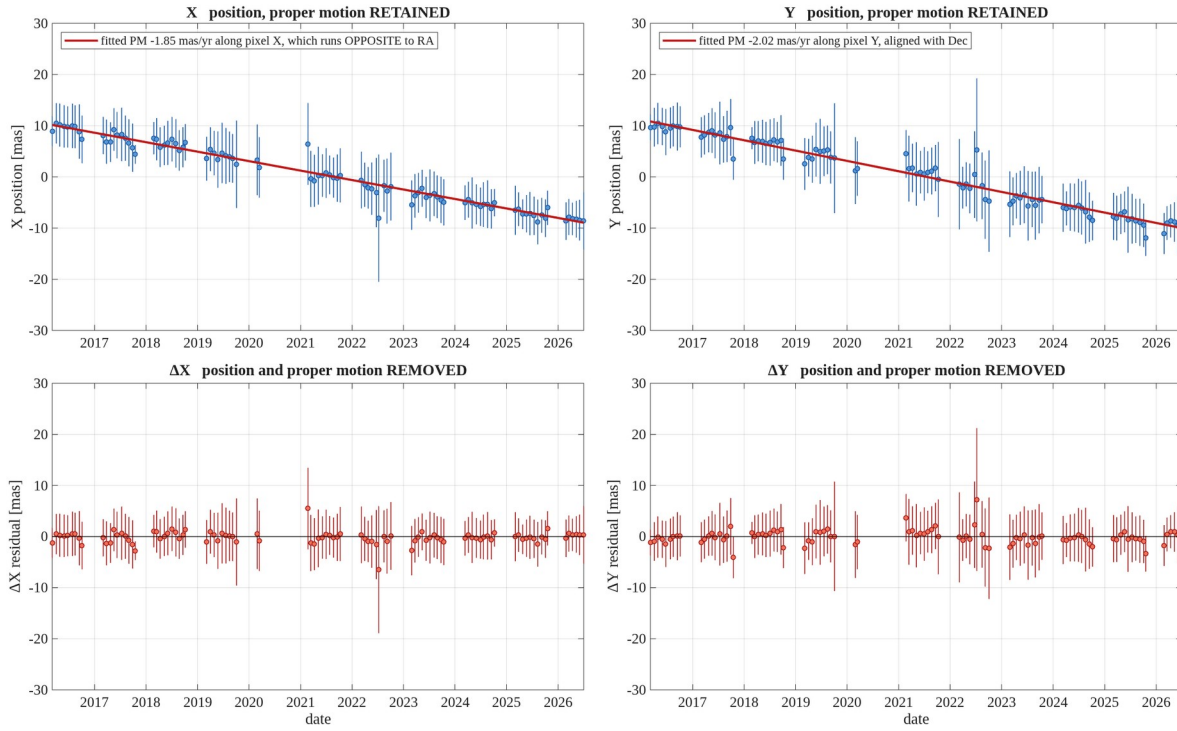


Figure 26. **v5, BLG01** — calibration star 43 pix from the target, freely fitted (v5 pins nothing). The same star as the corresponding v4 figure, but from the run in which **nothing is pinned** and Gaia enters only as a post-hoc tie. Sky motions quoted in the heading have the gauge removed and are absolute. Binned in one sidereal month; columns are the two axes, X left and Y right; top row keeps the proper motion, bottom row removes it. **Pixel X runs opposite to RA**, so the legend and the heading carry opposite signs in X.
report_v5_motion_BLG01_cal_d43.png

BLG01 comparison star outside the set, PM fitted — $l = 18.14$, 5 pix from the target, at (148,148), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \cos d / \mu_\delta$) $-0.71 / -5.42$ mas/yr

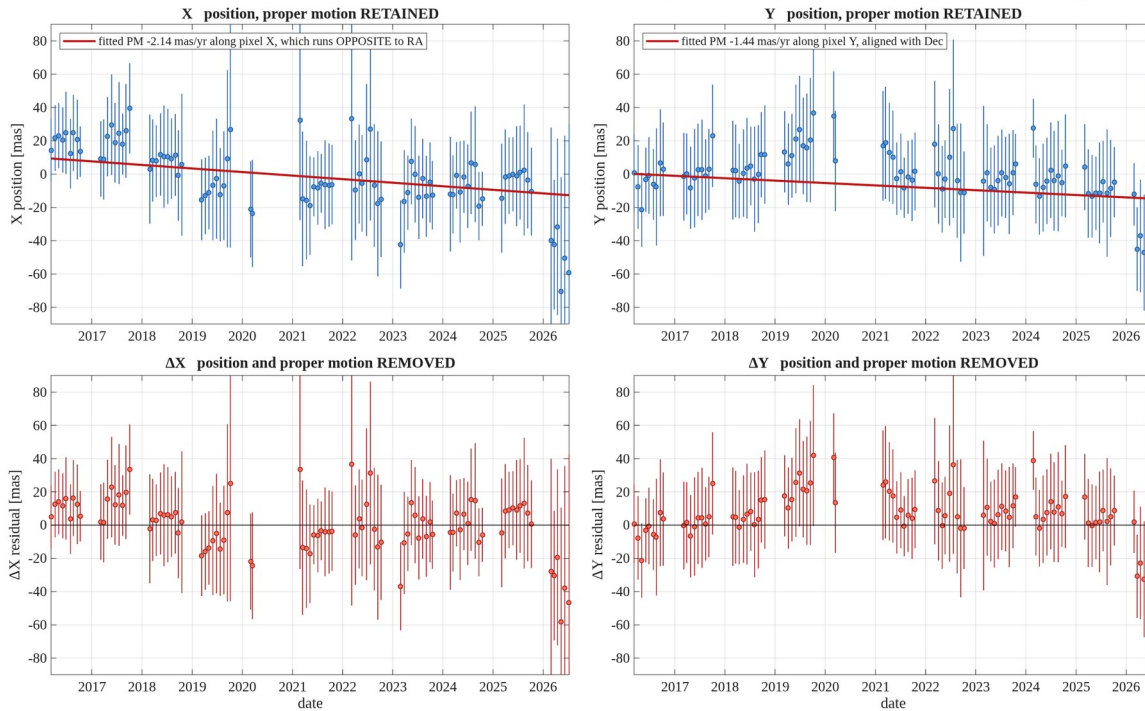


Figure 27. **v5, BLG01** — passenger at the target magnitude, 4.5 pix away — **blended with the target**, see section 7. The same star as the corresponding v4 figure, but from the run in which **nothing is pinned** and Gaia enters only as a post-hoc tie. Sky motions quoted in the heading have the gauge removed and are absolute. Binned in one sidereal month; columns are the two axes, X left and Y right; top row keeps the proper motion, bottom row removes it. **Pixel X runs opposite to RA**, so the legend and the heading carry opposite signs in X.
report_v5_motion_BLG01_out_d04.png

LG01 comparison star outside the set, PM fitted — $l = 17.91$, 21 pix from the target, at (156,171), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \cos \delta / \mu_\delta$) -8.49 / -9.74 mas/

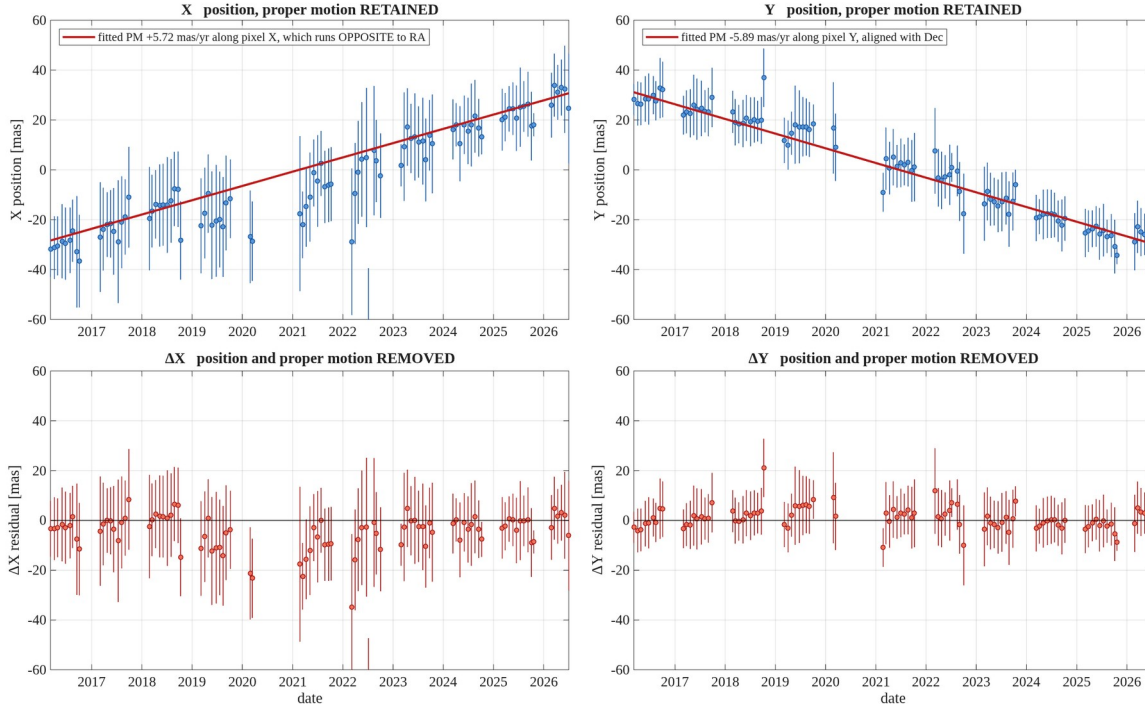


Figure 28. **v5, BLG01** — passenger at the target magnitude, 21 pix away, freely fitted. The same star as the corresponding v4 figure, but from the run in which **nothing is pinned** and Gaia enters only as a post-hoc tie. Sky motions quoted in the heading have the gauge removed and are absolute. Binned in one sidereal month; columns are the two axes, X left and Y right; top row keeps the proper motion, bottom row removes it. **Pixel X runs opposite to RA**, so the legend and the heading carry opposite signs in X. `report_v5_motion_BLG01_out_d21.png`

LG01 comparison star outside the set, PM fitted — $l = 18.05$, 49 pix from the target, at (200,145), freely fitted | ABSOLUTE sky PM ($\mu_\alpha \cos \delta / \mu_\delta$) +2.83 / -7.32 mas/

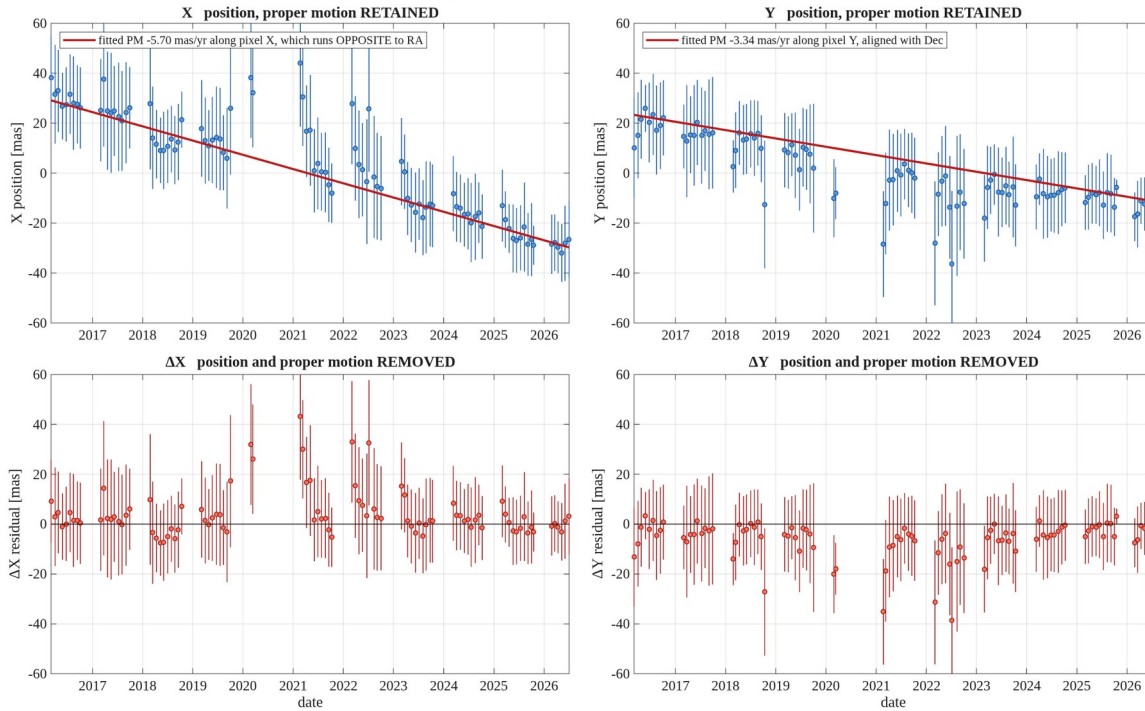


Figure 29. **v5, BLG01** — passenger at the target magnitude, 48 pix away, freely fitted. The same star as the corresponding v4 figure, but from the run in which **nothing is pinned** and Gaia enters only as a post-hoc tie. Sky motions quoted in the heading have the gauge removed and are absolute. Binned in one sidereal month; columns are the two axes, X left and Y right; top row keeps the proper motion, bottom row removes it. **Pixel X runs opposite to RA**, so the legend and the heading carry opposite signs in X. `report_v5_motion_BLG01_out_d48.png`